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PRICING THE UNCERTAIN: PREDICTION MARKETS AND QUANTITATIVE HEDGING UNDER EVENT RISK

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1 Abstract

This thesis develops and evaluates a systematic framework for incorporating event-linked prediction markets into quantitative hedging of financial exposures. Despite the theoretical appeal of prediction markets as aggregators of event-specific information, their use in risk management remains largely unexplored. The framework addresses this gap through a four-stage pipeline: principal-component factor residualization to isolate idiosyncratic asset returns, a bake-off-plus-validation model selection procedure that enforces strict temporal ordering throughout, an empirically calibrated branching-scenario simulation, and a dual-output hedge-sizing routine that reports both a point-estimate notional and a CVaR-augmented bootstrap notional. Each design choice is motivated by a specific failure mode — look-ahead bias, overfitting, equal-probability branch assumptions, and the conflation of PM-signal risk with innovation-distribution risk — that a naive implementation would overlook.

The framework is evaluated across three case studies that span materially different conditions: a conflict-linked commodity market, a political election market with time-varying salience, and a technology-expectations signal from non-cash forecasting platforms. The results reveal a structured gradient. In the Belgian electricity case, a regularized linear model assigns approximately 44% of its normalized importance weight to lagged prediction market returns, and a concrete hedge notional is triggered under both the point-estimate and bootstrap bases. In the Argentine electoral case, the framework correctly identifies that predictive structure is concentrated in a specific pre-election window rather than distributed across the full sample. In the AGI case, six independent diagnostics converge on a null result, and the framework produces a zero notional on the point-estimate basis — confirming that it correctly distinguishes genuine signal from noise. Taken together, the three cases define the conditions under which prediction-market-informed hedging is viable and those under which it is not, constituting both an empirical finding and a practical screening criterion for future applications.

The results carry direct implications for ongoing regulatory debates over the permissibility of prediction market contracts on conflict, terrorism, and other sensitive events. This thesis provides evidence that such contracts can serve legitimate risk-management functions for investors and traders with genuine event-linked exposures, and that a principled methodology for translating their informational content into hedging decisions now exists.

2 Introduction and Motivation

Prediction markets (PMs), sometimes also called information markets or decision markets, are organized markets in which participants trade contracts whose payoffs depend on the realization of a specified future event by a stated resolution date (Berg, Nelson, and Rietz 2008). A standard binary contract, for example, may pay \$1 if an event occurs and zero otherwise, so that its market price can be interpreted as a risk-adjusted assessment of the event’s likelihood. For example, if a prediction market security trades at 50 cents, there is a 50% chance of the event happening. In this sense, prediction markets transform uncertain future states of the world into tradable contingent claims. Their proposed applications are broad, ranging from forecasting and information aggregation to managerial decision support and institutional design (Hanson 2013; Justin Wolfers and Eric Zitzewitz 2004). At the same time, these markets are often viewed by the public and by regulators primarily through the lens of gambling (Paton, Siegel, and Vaughan Williams 2009). The tension between informational value and moral discomfort is central to understanding their current position in financial and political debates.

The recent growth of platforms such as Polymarket and Kalshi has made this tension much more visible. These platforms dominate retail-facing prediction market activity by trading volume (DeFi Rate 2024), and their rapid commercial expansion pushed event-based trading into mainstream discussion. As these markets have grown, so too has scrutiny over the types of events that should be tradable. A particularly important case for this thesis is the debate over contracts tied to violent conflict, terrorism, assassination, and war. In March 2026, U.S. legislators proposed the DEATH BETs Act, which sought to prohibit contracts referencing terrorism, assassination, war, or related activities (U.S. Congress 2026). Supporters of the bill argue that there is no public benefit in allowing markets that appear to profit from human suffering (Office of Representative Andy Levin and Schiff 2026). Related public pressure also contributed to the closure of Polymarket contracts related to nuclear detonation (Ostroff and Osipovich 2026). These developments show that PMs are not merely a niche forecasting technology; they are increasingly treated as politically and ethically consequential financial institutions, even if they are not yet systemically important.

This controversy matters analytically because the same contracts that appear morally contentious may also carry relevant information about consequential risks. The current discussion around PMs parallels longstanding debates in financial economics over short selling, speculative trading, and other practices that are often criticized on moral or stability grounds even when they contribute to price discovery and market completeness (Brunnermeier 2009; Beber and Pagano 2013). Controversy surrounding these markets further motivates this thesis and underscores the need to identify legitimate and empirically grounded use cases.

From a financial perspective, prediction market contracts are economically similar to other contingent claims whose payoffs depend on future events. This places them in close conceptual proximity to derivatives, insurance contracts, and catastrophe-linked securities, all of which are used to transfer risk across market participants (Hull 2018; Froot 2001; Ross 1976). In each case, an uncertain event is mapped into a payoff structure that can be priced, traded, and incorporated into a broader portfolio or risk-management framework. The financial economics literature has long emphasized that such instruments can serve two important functions: they help aggregate dispersed information, and they allow risks to

be reallocated toward agents more willing or able to bear them (Justin Wolfers and Eric Zitzewitz 2004). PMs are therefore potential instruments or information signals for hedging event-driven exposures.

This use case is especially relevant in settings where traditional financial assets are exposed to identifiable tradable events. If a security is sensitive to a specific event, then a market that prices that event should contain information useful for managing the event-driven risk. In principle, PMs may assist risk management by revealing updated market-implied probabilities about events that matter for asset returns. Despite this intuitive appeal, the use of PMs for quantitative hedging of financial exposures remains underexplored relative to their use for forecasting.

This thesis is motivated by that gap. It asks whether event-linked PMs can be incorporated into an empirical risk-management framework for traded securities. Put simply, the central question is: if an event can be linked to the returns of a security, can the PM associated with that event be used to assist with risk management?

3 Literature Review

3.1 Prediction Markets

3.1.1 Prediction Markets as Information Mechanisms

In his influential essay, “The Use of Knowledge in Society,” Hayek (1945) argued that markets efficiently aggregate the dispersed beliefs of their participants. The price of a security reflects investor expectations regarding its potential payoff. This idea extends naturally to prediction markets, where the price of a contract directly corresponds to the perceived likelihood of an event occurring (Manski 2006; Stark and Cortis 2017; Justin Wolfers and Eric Zitzewitz 2004). As such, PMs function as mechanisms for aggregating information and conveying collective expectations in an intuitive, continuously updated form.

A substantial body of experimental evidence supports the informational efficiency of these markets. In an early study of a political prediction market, Forsythe et al. (1992) found that “the market worked extremely well, dominating opinion polls. . . even though traders. . . exhibited substantial. . . judgment biases.” Similarly, Justin Wolfers and Eric Zitzewitz (2006) conclude that “prediction market prices typically provide useful (albeit sometimes biased) estimates of average beliefs.” These findings suggest that PMs can effectively aggregate dispersed information, even when the individual market participants exhibit systematic biases. However, this also implies that PMs may be affected by the cognitive biases of their participants.

To mitigate these issues, modern PMs often incorporate automated market-making mechanisms and proper scoring rules. For example, Hanson (2003) introduced logarithmic market scoring rules that incentivize truthful revelation of beliefs while maintaining liquidity. These mechanisms improve price formation and reduce the impact of thin trading, thereby enhancing the reliability of market-implied probabilities.

Justin Wolfers and Eric Zitzewitz (2004) extended the theoretical foundation by demonstrating how PMs can be designed to elicit not only point estimates, but also conditional forecasts and beliefs about entire distributions. This framework justifies the use of PMs to capture sentiment and quantify uncertainty. Snowberg, J. Wolfers, and E. Zitzewitz (2008) further highlighted the empirical relevance of PMs by showing how they can be used to extract macroeconomic signals, such as the impact of political events on economic expectations. Particularly relevant for this thesis, Aitken (2011) extend this application to the domain of security and terrorism, illustrating how PMs can reveal otherwise unobserved sources of geopolitical risk.

Indeed, a growing corpus of empirical literature demonstrates that PMs often perform as well as, or better than, traditional forecasting methods (Arrow et al. 2008; Berg, Nelson, and Rietz 2008). Taken together, these contributions underscore the value of PMs as tools for aggregating information and generating forward-looking signals about events.

3.1.2 Prediction Markets as Hedging Instruments

PMs have also been explored as tools for hedging event-driven risks, particularly in the context of political and regulatory uncertainty. Bergfjord (2008) examines their application in political risk management and argues that contracts tied to specific events—such as elections

or policy changes—can function as targeted hedges for firms exposed to these risks. In a similar vein, Goodell, McGee, and McGroarty (2020) provide empirical evidence linking prediction market probabilities to broader financial market dynamics. They show that changes in the incumbent party’s probability of winning an election explain a substantial portion of variation in financial market uncertainty, suggesting that prediction market prices contain information relevant for managing exposure to macroeconomic and policy-related risks.

Because they translate uncertain future events into tradable factors, PMs are conceptually appealing for hedging purposes. However, as discussed by Axén and Cortis (2020), the practical implementation of such strategies is subject to important limitations. In particular, transaction costs and the absence of “lay” positions in certain markets restrict the ability to form fully symmetric hedges, reducing their effectiveness in real-world settings.

Evaluating the performance of these hedging strategies requires careful empirical validation. Standard approaches from the trading literature, such as those outlined in Pardo, Serrano, and Fernández (2005), provide useful tools for simulation and backtesting, including performance measures like Sharpe ratios and drawdowns. At the same time, Bailey et al. (2014) highlight the risk of overfitting in backtest-based evaluations and propose statistical procedures to guard against false discoveries. Taken together, the literature suggests that while PMs offer a promising framework for hedging event risk, their effectiveness depends on both market structure and careful evaluation.

3.1.3 Limitations of Prediction Markets: Dynamics, Size, Scope, Regulation, and Data

Despite their conceptual appeal, PMs exhibit several structural limitations that constrain their effectiveness for trading and hedging applications. Most notably, these markets remain relatively small and illiquid, leading to significant challenges for trade execution. Low trading volumes and limited market depth result in wide bid–ask spreads and substantial slippage when executing even moderately sized positions. As noted by the think tank Panewslab (2023) and by Majumder et al. (2009), even prominent platforms such as Polymarket are vulnerable to large trades, allowing individual participants to exert disproportionate influence on prices. Empirical evidence supports this concern: approximately 63% of short-term contracts exhibit zero trading volume over a 24-hour period, and more than half of these contracts have less than \$100 in available liquidity.

These features imply that prices may not always reflect a well-aggregated consensus, particularly in thinly traded markets. As emphasized by Acadian Asset Management (2023), limited liquidity and shallow order books increase susceptibility to manipulation and execution inefficiencies. In practice, this means that even modest trades can “move the market,” undermining the feasibility of using these instruments as reliable hedging vehicles, consistent with the findings of Bergfjord (2008). Furthermore, low participation can lead to so-called “liquidity traps,” where prices are driven more by noise or automated trading activity than by new information. Whelan (2023) further highlights that transaction fees introduce a favorite–longshot bias, systematically distorting prices and reducing the reliability of market-implied probabilities. Taken together, these characteristics differentiate PMs from traditional, liquid derivative markets.

In addition to market microstructure constraints, the regulatory environment for PMs

varies considerably across jurisdictions, further limiting their accessibility and practical use. In the United States, platforms such as Kalshi operate legally under the oversight of the Commodity Futures Trading Commission (CFTC) as registered Designated Contract Markets. In contrast, Polymarket was fined \$1.4 million in 2022 for operating without proper registration and offering unregulated swaps to U.S. users, resulting in restrictions on its availability within the country (Marjanović 2025). In Europe, PMs are typically classified as gambling and are subject to corresponding legal constraints. While the United Kingdom permits limited operation under the Gambling Commission, several countries—including Belgium and France—prohibit such markets entirely. Similar restrictions apply across much of the Asia-Pacific region (Marjanović 2025). This fragmented regulatory landscape complicates cross-border participation and limits the scalability of PM-based strategies.

Despite these limitations, prediction market data remains widely accessible and well-curated, but with frequency limitations. Platforms such as Polymarket Analytics (2025b) aggregate and provide both live and historical data from major exchanges, including Polymarket and Kalshi, covering more than 60,000 markets with frequent updates. These datasets include detailed information on prices, trading volumes, open interest, and user positions, and are increasingly used in both academic research and media analysis. The reliability of these data sources is supported by their frequent citation in financial journalism outlets such as *The Wall Street Journal* and *CoinDesk* (Polymarket Analytics 2025a). Consequently, while execution and regulatory constraints limit the possibility of a pure prediction market hedge, the availability and quality of their data make them a valuable resource for empirical analysis.

3.2 Hedging and Risk Management

3.2.1 Risk-Neutral Valuation and the Foundations of Hedging

In a risk-neutral framework, the optimal hedge is defined as the position that maximizes the investor’s expected future wealth. V. Henderson (2005) describes this principle in the context of real option valuation, where the agent is assumed to choose the hedge that maximizes expected terminal wealth. Ziemba and Vickson (2003) further clarify that while expected wealth maximization is appropriate for risk-neutral agents, risk-averse investors typically seek to maximize the expected utility of wealth. Nonetheless, the expected value framework remains foundational for analyzing hedging strategies in quantitative finance. The notion of expected value itself is rigorously defined in probability theory as the weighted average of all possible outcomes, where the weights are given by the probabilities of those outcomes. Hansen (2022) provides a formal treatment of expectation in the context of random variables, emphasizing its central role in rational decision-making under uncertainty. This framework is directly relevant to hedging under uncertainty.

3.2.2 Hedging Using Binomial Trees

Hull (2000) presents a classical framework for optimal hedging in a binomial tree setting. By constructing a replicating portfolio that mimics the payoff of an option through a dynamically adjusted hedge ratio, it is possible to price derivatives as the discounted expected value of

their future payoffs under the risk-neutral measure. At each node in the binomial tree, the hedge ratio (Δ) is chosen to eliminate risk locally, ensuring that the portfolio evolves deterministically.

3.2.3 Scenario-Based Hedging and Downside Risk Measures

In practical applications, exact replication is rarely feasible, and hedging decisions are instead implemented using scenario-based frameworks. Expected losses are computed as probability-weighted outcomes across possible future states, often with a focus on downside realizations. This leads naturally to downside-oriented loss functionals, consistent with shortfall risk and tail-risk measures such as expected shortfall (CVaR), where losses are penalized asymmetrically relative to gains (Rockafellar and Uryasev 2000). The use of scenario-weighted loss aggregation is standard in simulation-based financial engineering and empirical risk analysis (Glasserman 2004; Ruppert and Matteson 2015).

3.2.4 Risk Budgeting and Partial Hedging Strategies

Within this framework, hedging is typically implemented relative to a predefined risk budget. Rather than eliminating all risk, investors hedge only the portion of expected loss that exceeds a tolerance threshold. This approach is central to risk budgeting in portfolio management, where exposures are scaled to align with acceptable risk levels rather than fully neutralized (Grinold and Kahn 2000; Meucci 2007). Operationally, this implies that no hedge is applied when expected losses remain within the risk budget, while hedge coverage increases as this threshold is exceeded. Such proportional coverage rules are consistent with theories of partial and efficient hedging under constraints, where full replication is suboptimal due to costs or capital limitations (Föllmer and Leukert 2000; Föllmer and Leukert 1999).

3.2.5 Market Frictions and Implementation Constraints

The translation from theoretical hedge ratios to implementable positions requires explicit consideration of execution. Transaction costs, market impact, and liquidity constraints reduce realized hedge performance relative to frictionless benchmarks, implying that hedge positions must be adjusted to account for implementation shortfall (Perold 1988; Almgren and Chriss 2001; Bertsimas and Lo 1998). In practice, this motivates the use of execution-aware adjustments such as haircut factors and staged execution strategies (e.g., time-weighted average price), which are commonly employed to manage trading costs (Kissell 2014).

In addition, hedging effectiveness is typically less than one due to basis risk, path dependence, and imperfect correlation between the hedge instrument and the underlying exposure. As discussed in Hull (2018), real-world hedges rarely achieve perfect replication, motivating the use of conservative hedge-efficiency parameters when scaling hedge positions. Even when explicit modeling of implementation costs is not feasible, conservative execution adjustments remain appropriate given that such costs can be material.

3.2.6 Hedging Instruments and Payoff Structures

The choice of hedging instruments depends on both the magnitude and structure of downside risk. Option-based overlays provide a flexible set of tools with distinct payoff profiles, including protective puts, collars, and put spreads. These instruments are widely used in institutional portfolios and involve well-documented trade-offs between cost, convexity, and upside participation (Israelov and Nielsen 2014; Clarke, Silva, and Thorley 2011). For example, put spreads reduce premium costs at the expense of limited protection, while collars finance downside protection through call writing, thereby sacrificing some upside potential.

In practice, these instruments are often deployed through severity-based policy rules that map downside risk levels to specific hedge structures. Mild downside scenarios are typically addressed using lower-cost instruments such as out-of-the-money put spreads, while more severe risks justify the use of protective puts or collars. In extreme scenarios, hedging may be complemented by partial de-risking (selling or shorting) or broader market overlays, reflecting the importance of tail-risk mitigation and capital preservation. This approach aligns with dynamic hedging frameworks that emphasize state-dependent strategies under uncertainty and non-linear payoffs (Taleb 1997).

3.3 Methodological Literature

3.3.1 Factor-model decomposition

Decomposing asset returns into systematic and idiosyncratic components is central to our methodology. Factor models provide a natural starting point for this task. Following the classical asset-pricing literature, returns can be represented as a function of common risk factors together with an idiosyncratic residual that captures variation not explained by broad market movements (Fama and French 1993). This decomposition is widely used in both academic and practitioner settings to isolate economically meaningful signals and to control for confounding sources of variation (Grinold and Kahn 2000). More recent practitioner-oriented textbooks emphasize the role of factor models in modern quantitative investment workflows, particularly for risk decomposition and alpha identification (Paleologo 2021; Paleologo 2025). In our setting, this step is essential because it allows the analysis to focus on the portion of returns that may plausibly be linked to event-specific information from PMs, rather than to general market or sectoral comovement.

3.3.2 Traditional Time-Series Benchmarks

After isolating the return component of interest, we consider a set of traditional time-series models to describe the dynamic relationship between asset returns and prediction market returns. Standard references such as Tsay (2005) and Shumway and Stoffer (2017) outline a broad class of models for financial time series. In univariate settings, Autoregressive Integrated Moving Average (ARIMA) models are useful for capturing persistence and handling nonstationarity. In multivariate settings, Vector Autoregression (VAR) models provide a flexible framework in which each variable is modeled as a function of its own lags and the lags of the other variables in the system (Sims 1980a). For this application, VAR models are

attractive for data exploration because they explicitly represent the joint evolution of asset returns and event-linked market variables.

To move from reduced-form dependence to economically interpretable shocks, the VAR framework can be extended to a Structural VAR (SVAR). This requires an identification scheme, because reduced-form innovations are generally contemporaneously correlated and therefore cannot be interpreted directly as distinct economic shocks (Rubio-Ramirez, Waggoner, and Zha 2010). In practice, several strategies are commonly used for structural identification. Recursive identification based on a Cholesky decomposition imposes short-run ordering assumptions, while sign-restricted SVARs use economically motivated restrictions on the direction of impulse responses and are therefore often viewed as more flexible in macro-finance settings (Uhlig 2005; Fry and Pagan 2011). External-instrument, or proxy, SVAR methods offer an even more causal interpretation when valid instruments are available (K. Mertens and Ravn 2013; Stock and Watson 2018).

As a robustness check for dynamic effects, local projection methods may also be employed. Rather than imposing the full dynamic structure of a VAR, local projections estimate horizon-specific responses directly and are therefore less sensitive to certain forms of model misspecification (Jordà 2005). In addition, because financial returns often exhibit volatility clustering, Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models can be combined with ARIMA- or VAR-type specifications to allow for time-varying conditional variances and more realistic simulated paths (Tsay 2005; Mandelbrot and Hudson 2004).

3.3.3 Machine-Learning Candidates

In addition to traditional econometric models, we consider a range of machine-learning methods for event-linked return prediction. In this setting, the objective is not only to characterize dependence over time, but also to exploit potentially nonlinear, weak, and high-dimensional relationships between asset returns and prediction market returns. Machine-learning models are well suited to this task because they can combine lagged variables, rolling features, and event-specific covariates in a flexible predictive framework. Regularized linear models such as ridge, lasso, and elastic net are useful when the predictor set is large and highly collinear (Hoerl and Kennard 1970; Tibshirani 1996; Zou and Hastie 2005). Tree-based methods such as random forests, extremely randomized trees, and gradient boosting allow for nonlinearities and interactions without requiring them to be specified *ex ante* (Breiman 2001a; Geurts, Ernst, and Wehenkel 2006; Friedman 2001). Support vector regression provides an additional class of flexible estimators, particularly when nonlinear kernels are used (Smola and Schölkopf 2004).

Because the data are time ordered, these models must be estimated and tuned using time-respecting validation procedures rather than standard i.i.d. cross-validation. Rolling-origin, expanding-window, or blocked-fold validation preserves temporal ordering and helps prevent look-ahead bias in model selection and performance evaluation (Bergmeir and Benítez 2012; Hyndman and Athanasopoulos 2021). This is particularly important in financial applications, where the predictive signal is often weak and highly sensitive to leakage. The detailed machine-learning models considered in this thesis are described below.

Ridge regression. Hoerl and Kennard (1970) introduce ridge regression, which fits a linear model by minimizing squared error subject to an ℓ_2 penalty on coefficients; the penalty shrinks estimates toward zero to stabilize inference under multicollinearity. In time-series forecasting (frequency, horizon, and sample size unspecified), ridge is often a strong baseline when many correlated predictors are used (e.g., lags, rolling statistics, and prediction market features), because shrinkage reduces variance and mitigates noise-fitting in asset returns. Its main limitation is the linear, time-invariant relationship it imposes: it may miss nonlinear or regime-dependent responses that are plausible when event probabilities approach 0/1 or when markets react discontinuously to news. In practice, predictors should be standardized, and the penalty λ should be selected using time-respecting validation; interpretability remains high (coefficient magnitudes), but ridge does not perform variable selection.

Lasso. The lasso of Tibshirani (1996) replaces the ℓ_2 penalty with an ℓ_1 penalty, yielding sparse models by setting some coefficients exactly to zero. This sparsity can be attractive for event-linked return modeling because it produces a small set of selected predictors (e.g., a subset of prediction market lags or interaction features), which can improve interpretability and reduce estimation error when signal-to-noise is low, as is common for asset returns. However, with strongly correlated time-series predictors (overlapping rolling windows, correlated contracts, or multiple lags), the lasso can be unstable, arbitrarily selecting one of many near-equivalent variables and changing selections across folds. Standardization is essential; the regularization strength should be tuned under rolling/blocked CV, and it is often useful to add time-to-event or regime indicators explicitly because the linear form cannot discover them automatically. Overfitting risk is lower than unpenalized regression, but selection-induced instability is a primary practical limitation.

ElasticNet. Zou and Hastie (2005) propose ElasticNet, which combines ℓ_1 and ℓ_2 penalties to balance sparsity with stability, encouraging a "grouping effect" that tends to keep correlated predictors together. For this use case, this specification is likely preferable to pure lasso if features are highly collinear (e.g., many lags of the same signal). ElasticNet retains the linearity assumption and therefore may underfit nonlinear responses (for example, regime switching), but it can be more robust than lasso when the predictive signal is diffuse across correlated variables. In practice, we aim to tune both the mixing parameter (relative ℓ_1 vs. ℓ_2 weight) and overall penalty with time-aware validation, and standardize predictors; interpretability is moderate (sparse-ish coefficients) and overfitting is controlled primarily through regularization.

Random Forest. Breiman (2001a) introduce Random Forests, which average many decision trees trained on bootstrap samples while randomizing feature selection at each split to reduce correlation among trees. Random forests capture nonlinearities and interactions without requiring explicit specification, which can be useful when prediction market variables affect returns in state-dependent ways (e.g., different sensitivities at different probability levels). For asset returns, however, the flexibility can be a liability: if the true predictive signal is weak and nonstationary, forests may learn spurious patterns unless leakage is tightly controlled and depth/leaf sizes are constrained. For prediction market returns (often bounded

and driven by discrete information arrivals), forests can fit nonlinear mappings but may struggle with abrupt jumps or structural breaks unless time-to-resolution and other time-aware features are included. Key hyperparameters include number of trees, maximum depth, minimum leaf size, and max-features; interpretability is limited to variable importance and partial dependence summaries, and overfitting risk is primarily controlled by restricting tree complexity and validating with proper time-series folds.

ExtraTrees. Extremely randomized trees of Geurts, Ernst, and Wehenkel (2006) modify tree ensembles by injecting additional randomness: splits may be chosen using randomly drawn cut points, often without bootstrap resampling. This typically reduces variance (and can speed training), which is helpful in noisy return prediction problems where overfitting is a central concern. Relative to Random Forests, ExtraTrees may introduce more bias, so they can underperform when the signal is genuinely complex but may be more stable when relationships drift over time. For event-linked applications, ExtraTrees benefit from explicit time-aware predictors (e.g., time to event resolution) because the algorithm itself is not time-aware and cannot prevent look-ahead. Practical tuning focuses on number of trees and controlling tree depth/leaf size; interpretability and limitations are similar to Random Forests.

Gradient Boosting. Friedman (2001) formalize gradient boosting as stagewise additive modeling: weak learners (typically shallow trees) are fit sequentially to reduce a specified loss, producing a flexible function approximator. Boosting is often highly competitive on tabular forecasting problems because it captures nonlinearities and interactions while allowing careful regularization through shrinkage (learning rate), tree depth, and the number of boosting iterations. In return forecasting, this power can overfit quickly if the horizon is short and the signal is weak, so early stopping (via time-series validation) and conservative depth are crucial. For prediction market returns, boosting can model nonlinear responses to probability levels and contract-specific features, but it remains a reduced-form predictor and will not distinguish causal from spurious relationships without additional design. Key hyperparameters include learning rate, number of estimators, max depth, subsampling, and minimum leaf size; interpretability is improved via feature importance and partial dependence diagnostics, but models remain less transparent than linear methods.

SVR (RBF kernel). Support Vector Regression extends margin-based learning to real-valued prediction; Cortes and Vapnik (1995) provide the foundational support-vector framework and Smola and Schölkopf (2004) detail SVR for function estimation with ε -insensitive loss and kernel-based nonlinear feature maps. With an RBF kernel, SVR can fit smooth nonlinear relationships, which can be useful when mapping prediction market signals into asset returns exhibits nonlinear saturation or interaction effects. However, SVR requires careful preprocessing and tuning: predictors should be standardized, and performance is sensitive to (C, γ, ε) ; poorly tuned kernels can either underfit (too smooth) or overfit (too wiggly), especially under nonstationarity. Computational cost can become limiting as sample size grows, and interpretability is typically low because predictions depend on support vectors rather than explicit coefficients. In practice, SVR is best treated as a medium-sample nonlinear

baseline with conservative hyperparameter search under rolling/blocked CV.

SVR.linear. Linear SVR (Smola and Schölkopf (2004)) uses the same ε -insensitive loss but with a linear kernel, producing a linear predictor that is conceptually closer to penalized regression while retaining robustness features tied to margin-based estimation. This can be attractive for event-linked return modeling when the goal is a stable linear mapping from prediction market features to returns, but with reduced sensitivity to small residual variations. As with ridge/lasso, the model assumes a time-invariant linear relationship and therefore may miss nonlinear threshold behavior common near event resolution; nonetheless, it can be competitive when the predictive signal is weak and sparse. Standardization is important, and key hyperparameters are C (regularization strength) and ε (insensitivity band), both of which should be tuned under time-respecting validation. Interpretability is moderate (linear weights), and overfitting is controlled via C and feature design rather than tree complexity.

FoundationModel (tuned GradientBoosting variant). In this thesis, the "FoundationModel" refers to a strongly tuned gradient-boosted tree system—not a pretrained transformer—motivated by the empirical dominance of boosted trees on many structured prediction tasks (Friedman 2001; Chen and Guestrin 2016). Conceptually, it is gradient boosting with more aggressive regularization and hyperparameter optimization (e.g., shrinkage, depth constraints, subsampling/column sampling, and explicit penalties), aiming to serve as a high-performance baseline for event-linked return prediction. The main strength is predictive flexibility: it can learn nonlinearities between prediction market features and returns and can leverage engineered time-aware predictors (e.g., time to event resolution) to partially adapt to nonstationarity. The principal limitation is that it remains a reduced-form predictor and is prone to overfitting under regime change or weak-signal settings unless early stopping and careful time-series validation are enforced; interpretability is limited but can be supported with feature importance and partial dependence checks. In practice, use time-decay weights or rolling recalibration to address nonstationarity, and prefer conservative depth with sufficient regularization when forecasting asset returns.

Time-series cross-validation and evaluation. Standard i.i.d. k -fold cross-validation is generally invalid for time-series forecasting because random splits leak information across time, violating the temporal ordering and producing optimistically biased performance estimates (Bergmeir and Benítez 2012). A principled alternative is rolling-origin (expanding-window or sliding-window) evaluation, where each fold trains on observations up to time t and tests on a future block at the forecast horizon of interest (unspecified here); blocked CV is similar but uses contiguous train/test blocks and can include a "gap" to reduce leakage from overlapping labels or features (Hyndman and Athanasopoulos 2021). Hyperparameters should be tuned inside the rolling/blocked procedure (ideally nested) to avoid look-ahead. Report point-forecast metrics such as MAE and RMSE; use MAPE cautiously for returns near zero (or consider scaled alternatives), and report directional accuracy (e.g., sign correctness) with appropriate testing when relevant (Pesaran and Timmermann 1992).

3.3.4 Comparison of Model Classes

The traditional and machine-learning approaches considered here serve related but ultimately distinct purposes. Traditional time-series models such as VAR are especially valuable when the objective is structural interpretation. Their impulse response functions are derived from a well-defined dynamic system (Sims 1980b; Kilian and Lütkepohl 2017), making them particularly attractive for counterfactual analysis. However, they also rely on stronger assumptions including linear dynamics and relatively stable dependence structures.

By contrast, machine-learning models are designed primarily for predictive performance. In our setting, this flexibility is an important advantage. The relationship between asset returns and prediction market returns may be nonlinear, sparse, and state dependent, especially around salient events or near contract resolution. Regularized linear models help control overfitting when many correlated predictors are available, while tree-based models and boosting methods can capture interactions and threshold effects that traditional linear systems may miss (Breiman 2001b; Athey and Imbens 2017). Recent foundation-style forecasting approaches likewise emphasize predictive transfer and flexible reduced-form learning rather than structural identification (Ansari et al. 2024; Das et al. 2023). Their main drawbacks are reduced interpretability, sensitivity to regime shifts, and the fact that they do not, by default, deliver identified shocks or structural impulse responses.

For the purposes of this thesis, the main objective is accurate event-linked prediction and practical hedge construction rather than structural policy counterfactual analysis. For that reason, the advantages of machine-learning models are especially relevant. They allow richer functional forms, adapt more naturally to engineered time-series features, and are well suited to a rolling-validation framework aimed at out-of-sample performance. Although this comes at the cost of weaker structural interpretation, that trade-off is acceptable in the present application. Consequently, while traditional time-series models remain useful benchmarks and provide an important point of methodological comparison, the final empirical specification is based on a machine-learning model.

4 Methodology

4.1 Model Formulation

The goal of the modeling stage is to estimate how a security responds to changes in event expectations as reflected in a related PM. Rather than modeling only the final binary outcome, the empirical framework treats the PM as a time-varying signal whose returns summarize continuously updated beliefs about the event’s likelihood. The central object of interest is therefore the dynamic relationship between the return of the security and the return of the associated PM contract.

Let r_t denote the return of the asset at time t , let x_t denote the prediction market return, and let TTR_t denote the number of calendar days remaining until contract resolution. For the Belgian electricity case, the contract “Nuclear weapon detonation in 2025?” resolves on 2026-01-01, so TTR_t counts down from approximately 365 at the start of 2025 to 0 at year-end. Including time-to-resolution allows the model to represent the well-documented tendency for prediction market prices to carry increasing informational weight as the resolution date approaches. The modeling problem is to estimate the conditional evolution of the asset return, or of an asset-specific residual return, as a function of its own recent history, the recent history of the PM, and the remaining contract life:

$$r_t^* = f(r_{t-1}^*, \dots, r_{t-p}^*, x_{t-1}, \dots, x_{t-p}, TTR_t) + u_t,$$

where r_t^* denotes either the raw return or a factor-adjusted return, $f(\cdot)$ represents the forecasting model, and u_t is an innovation term. This formulation accommodates both traditional time-series models and machine-learning models. The resulting estimated relationship is used in later stages to generate scenario-based forecasts, evaluate sensitivity to event-related shocks, and inform hedge design.

4.1.1 Factor Model

To identify the sensitivity of the financial asset to changes in the PM, we first employ a factor-model framework to decompose observed returns into systematic and idiosyncratic components:

$$r_t = \beta' f_t + \epsilon_t,$$

where f_t denotes a vector of systematic factors and ϵ_t denotes the idiosyncratic return. This decomposition is standard in both academic and practitioner literature (Fama and French 1993; Grinold and Kahn 2000; Paleologo 2021; Paleologo 2025). Without filtering systematic effects, any apparent association between the asset and the PM could simply reflect shared exposure to common underlying drivers. By estimating the factor component and retaining the residual $\epsilon_t = r_t - \hat{\beta}' f_t$, we obtain a series more directly interpretable as an event-linked response.

When the candidate factor set is large or highly collinear, the model incorporates a dimension-reduction step: correlated explanatory variables are summarized through principal components (Jolliffe 2002), with the number of retained components chosen so that cumulative explained variance exceeds 95%. To prevent look-ahead bias, the singular value decomposition (SVD) used for this step is fitted exclusively on a held-back training window.

Specifically, the last 20 observations of the available return series are withheld as a buffer, and the SVD is estimated on the remaining $n_{\text{train}} = \max(30, n - 20)$ observations. The resulting factor loadings $V_{[:,k]}^\top$ are then fixed and applied to the full sample — training and held-out periods alike — so that no future information enters the factor construction step (Bergmeir and Benítez 2012; Hyndman and Athanasopoulos 2021). Estimating PCA on the full sample before carving out any holdout would allow covariance information from the withheld observations to leak back into the factor scores used during training, artificially inflating apparent model fit. For the Belgian electricity case with 363 daily returns this yields $n_{\text{train}} = 343$ factor-model training observations, retaining a small final buffer for diagnostics. Asset returns are expressed using a symmetric transform,

$$r_t = 200 \cdot \frac{p_t - p_{t-1}}{|p_t| + |p_{t-1}|},$$

which avoids the near-zero denominator instability of standard percentage changes when prices are close to or below zero, as occasionally observed in European day-ahead electricity markets. The factor mapping is estimated by Huber robust regression with $\epsilon = 1.35$, corresponding approximately to the 85th percentile of the standard normal, so that extreme price-spike observations are down-weighted rather than allowed to dominate the fitted coefficients (Huber 1964). The resulting idiosyncratic return series ϵ_t serves as the target variable in all downstream modeling.

4.1.2 Traditional time-series models

After isolating the return component of interest, we consider a set of traditional time-series models as benchmark specifications. The primary traditional benchmark is a reduced-form Vector Autoregression (VAR), in which the factor-adjusted asset return and the prediction market return are modeled jointly as functions of their own lagged values and the lagged values of the other series (Sims 1980a). Time-to-resolution (TTR_t) enters the VAR as a deterministic exogenous regressor — passed via the `exog=` argument of `statsmodels VAR.fit()` and supplied as `exog_future=` at forecast time — rather than as an endogenous variable. This means that the structural VAR block remains 2×2 (asset return and prediction market return), and TTR_t does not appear in the impulse responses or the identification step; it serves purely as a time-varying intercept shift that captures the well-documented tendency for prediction market informativeness to increase as the resolution date approaches. Lag order is selected by iterating over candidate lengths from 1 to $\min(10, \lfloor N/10 \rfloor)$ and retaining the lag that minimises holdout RMSE on a short validation slice carved from the training window, consistent with the prediction-error criterion used throughout the comparison.

More elaborate structural identification strategies for the VAR — including sign-restricted SVARs and external-instrument approaches — are discussed in the literature review as potential extensions, but are not implemented in the empirical comparison (Uhlig 2005; Fry and Pagan 2011; K. Mertens and Ravn 2013; Stock and Watson 2018).

Finally, GARCH-type specifications allow the conditional variance of returns to evolve dynamically, improving the representation of volatility clustering and the realism of simulated paths (Tsay 2005; Mandelbrot and Hudson 2004). The GARCH model is estimated by maximum likelihood, with the ARMA order of the conditional mean equation selected by BIC prior to fitting the variance equation.

4.1.3 Machine-learning time-series models

In addition to traditional econometric models, we consider a range of machine-learning methods for event-linked return prediction. The primary objective is predictive accuracy rather than structural identification, which is especially relevant because the relationship between asset and prediction market returns may be weak, nonlinear, and state dependent. The feature set at each observation consists of $p = 3$ lags of both the idiosyncratic asset return and the prediction market return, together with TTR_t , giving seven predictors in total.

Regularized linear methods—Ridge, Lasso, and ElasticNet—are useful when the predictor set is collinear (Hoerl and Kennard 1970; Tibshirani 1996; Zou and Hastie 2005). All three, together with both SVR specifications, are estimated inside a **StandardScaler** pipeline so that unit differences across the lagged series do not distort the distance metric or the regularization penalty. Tree-based ensemble methods—RandomForest, ExtraTrees, and GradientBoosting—do not require scaling as they are invariant to monotone feature transformations (Breiman 2001a; Geurts, Ernst, and Wehenkel 2006; Friedman 2001). Support vector regression with an RBF kernel provides a further nonlinear alternative (Smola and Schölkopf 2004). A strongly tuned boosted-tree variant labeled *FoundationModel* completes the candidate set; it refers not to a pretrained transformer but to a gradient-boosting system with aggressive regularization motivated by the empirical dominance of boosted trees on structured prediction tasks (Friedman 2001; Chen and Guestrin 2016).

Model estimation and hyperparameter tuning are carried out using time-respecting validation procedures throughout. Rolling-origin and expanding-window procedures preserve temporal ordering and reduce the risk of look-ahead bias (Bergmeir and Benítez 2012; Hyndman and Athanasopoulos 2021).

4.1.4 Comparison of model classes and final specification

The empirical comparison involves ten candidate specifications: VAR as the traditional benchmark; Ridge, Lasso, ElasticNet, SVR with an RBF kernel, linear SVR, RandomForest, ExtraTrees, GradientBoosting, and FoundationModel as the machine-learning class; and a GARCH specification. Selecting among these requires a common performance criterion valid across all eleven models simultaneously. Information criteria such as AIC and BIC cannot serve this role because they require a well-defined likelihood function and a fixed, countable parameter set. Both conditions hold for the Gaussian VAR and GARCH (Tsay 2005; Shumway and Stoffer 2017). They do not hold for SVR, whose regularization penalty has no direct likelihood interpretation, nor for tree-based ensembles, which optimize a loss function over a combinatorial parameter space with no equivalent degrees-of-freedom count (Breiman 2001a; Geurts, Ernst, and Wehenkel 2006; Friedman 2001). Penalized-likelihood variants of AIC and BIC exist for Ridge and Lasso in the univariate regression literature, but their extension to lagged multivariate settings with cross-validated tuning parameters is non-standard. A single prediction-error criterion requiring only point forecasts and realized outcomes is therefore used throughout.

The comparison proceeds in two stages with distinct training-window rules that serve different purposes.

In the first stage, all eleven models are evaluated on a common fixed holdout split: the

last 10 observations of the full sample are withheld as a bakeoff test set, and each model is trained on the remaining observations. The minimum training requirement for this stage is $\max(50, n_{\text{lags}} + 20)$ observations, which ensures enough data for stable estimation before the fixed holdout is evaluated. RMSE and MAE are the primary metrics: RMSE penalizes large errors heavily, making it sensitive to the occasional large spike in the idiosyncratic return series, while MAE is more robust to those extreme observations. Directional accuracy—the proportion of one-step-ahead forecasts with the correct sign—is reported as a secondary metric, since a model that systematically predicts the wrong direction would be misleading regardless of its RMSE.

In the second stage, shortlisted models are evaluated under both expanding-window and rolling-window forward-looking splits. Here the minimum training set uses the adaptive formula $\max(30, \lfloor n_{\text{obs}}/3 \rfloor)$, which serves a different goal from the bakeoff minimum: it ensures that at least one-third of the available data is always used for training regardless of sample length, avoiding the pathological case in which a hardcoded floor would consume the entire dataset when the series is short (as with weekly or otherwise lower-frequency energy data). The window steps forward by a stride of $\max(5, \lfloor n/20 \rfloor)$ observations at each origin and is capped at 80 forecast origins. The rolling window uses the same step and test-block configuration with a fixed window length of $\max(30, \lfloor n/3 \rfloor)$. The composite validation rank is formed as the sum of four equally weighted sub-ranks: mean RMSE under expanding validation, mean RMSE under rolling validation, directional accuracy under expanding validation, and directional accuracy under rolling validation. The final model is selected by summing the first-stage bakeoff rank and the composite validation rank, giving both stages equal weight. The selected model is then used exclusively for simulation and hedge sizing.

Feature importances are extracted using the following hierarchy: impurity-based feature importances for tree-based models; absolute coefficient values $|\hat{w}_j|$ for linear models; and permutation importance for SVR (Altmann et al. 2010), which measures the increase in RMSE when each feature is randomly permuted. For linear models (Ridge, Lasso, ElasticNet), the absolute coefficients are properties of the model fitted on the full training window and do not depend on a holdout split; the 80/20 split described below applies only to the permutation importance procedure used for SVR. For permutation importance, the estimator is fitted on the first 80% of the training window, and importance scores are computed on the held-out 20% of that window; this holdout-based evaluation prevents self-reporting bias that would arise if the same observations were used for both fitting and importance assessment. Raw absolute coefficients for linear models do not sum to 1.0 by construction (unlike impurity-based tree importances), so they are normalized by dividing by their absolute sum before reporting, ensuring shares are comparable across features and cases. When the training window is too short (fewer than 25 observations in total), the procedure falls back to in-sample evaluation. Permutation importance estimates are noisier than impurity-based importances, particularly on short samples, and are interpreted accordingly.

4.1.5 Scenario-Based Branching Simulation

Once the forecasting model has been selected, it is used to generate a compact branching set of forward scenarios. The shocks that define the branches are drawn from the empirical distribution of observed prediction market returns: the 25th percentile (q_{down}) and 75th

percentile (q_{up}) of the historical PM return series serve as representative negative and positive shocks respectively. Using the interquartile range produces branches that represent typical rather than extreme market movements, which is appropriate given that a one-year daily series provides limited support for tail quantile estimation.

Formally, let x_t denote the prediction market return and let $\hat{f}(\cdot)$ denote the selected predictive model for the idiosyncratic return ϵ_t . Starting from the most recent observed history, we simulate forward responses over a horizon of $H = 5$ days. The PM return path for a given branch is constructed as a sparse impulse: the stage-1 shock q_{up} or q_{down} is applied at step 1, the stage-2 shock is applied at step 2, and the PM return is set to zero at all subsequent steps ($h = 3, 4, 5$). Concretely, for each branch (s_1, s_2) the exogenous PM return path is:

$$x_{t+h}^{(s_1, s_2)} = \begin{cases} q_{s_1} & h = 1, \\ q_{s_2} & h = 2, \\ 0 & h = 3, 4, 5, \end{cases}$$

so that the branching shocks represent a transient two-period impulse rather than a persistent shift in the PM return process. The model produces a corresponding path of predicted abnormal returns,

$$\hat{\epsilon}_{t+h}^{(s)} = \hat{f}\left(\mathcal{H}_{t+h-1}^{(s)}\right),$$

where $\mathcal{H}_{t+h-1}^{(s)}$ denotes the lagged history of both series under scenario s . The cumulative abnormal return (CAR) at each terminal node therefore reflects how strongly the model propagates the initial two-step shock through the autoregressive lag structure, not the effect of a sustained change in geopolitical risk expectations. This design is intentionally conservative: by allowing PM returns to revert to zero after the two-stage event, the framework avoids extrapolating a persistent shock beyond what the short historical sample can support. A two-stage tree is used, yielding four terminal paths. Branch probabilities are not assumed equal; instead, they are calibrated from historical data. Conditional transition probabilities — $P(\text{up} \mid \text{up})$, $P(\text{down} \mid \text{up})$, $P(\text{up} \mid \text{down})$, and $P(\text{down} \mid \text{down})$ — are estimated empirically from the observed sequence of up and down moves in the prediction market return series. To avoid zero-probability branches when one transition type has never been observed in the available history, Laplace smoothing (additive smoothing with pseudocount 1) is applied to each conditional count before normalizing to probabilities — a technique that adds a small positive count to each unobserved transition category so that no empirical probability is exactly zero, at the cost of a modest pull toward uniform probabilities (Goodman 2001). This prevents degenerate trees in which a branch carries zero weight and would otherwise be invisible to the hedge-sizing calculation. The terminal quantity of interest is the cumulative abnormal return (CAR) implied by each path.

The tree design generalises to K stages and N sub-branches depending on the nature of the prediction market contract. For binary event contracts—where the PM price represents the probability of a discrete outcome (win/lose, detonate/not)—a two-stage, four-branch design is used, with up/down transition probabilities calibrated from the empirical sequence of daily PM returns. For continuous-scale contracts where the PM signal represents a quantity on a real-valued scale (such as an AGI-timeline measured in years), a three-stage, nine-branch design is used instead, with stage-1 regime probabilities set to the 10th, 50th, and 90th

percentiles of the empirical distribution of observed multi-step shifts; within each regime, stage-2 sub-branches use the same quantile calibration. The two calibration procedures differ because the up/down transition framework is not meaningful for a continuously-valued signal, while the quantile-based regime design is not appropriate for a binary contract where the terminal resolution is a discrete 0/1 event. Both designs share the same impulse-path structure (shocks at steps 1–2, zero thereafter) and the same CAR accumulation and CVaR augmentation framework.

This scenario design is intentionally parsimonious: it does not characterize the full distribution of future prices but provides a tractable way to study how the model reacts to plausible PM shocks, consistent with the broader simulation logic in financial engineering (Glasserman 2004; Ruppert and Matteson 2015) and the emphasis on state-dependent risk management in dynamic hedging (Taleb 1997).

4.1.6 Scenario-Based Loss Assessment and Hedge Sizing

Point-estimate loss and hedge coverage. The scenario tree is used to summarize downside exposure under alternative prediction market realisations. For each terminal scenario s , let r_s denote the CAR and let w_s denote the scenario probability. Losses are defined asymmetrically according to the exposure convention. A *long-price* participant (e.g. a producer who benefits from higher prices) is harmed by falling prices, so $L_s = \max(-r_s, 0)$. A *short-price* participant (e.g. an energy consumer who benefits from lower prices) is harmed by rising prices, so $L_s = \max(r_s, 0)$. The appropriate branch is selected based on the stated exposure convention before the probability-weighted loss is computed. The expected downside loss is

$$\bar{L} = \sum_s w_s L_s.$$

This asymmetric construction reflects the fact that hedging is driven by downside risk rather than symmetric forecast error, connecting naturally to shortfall-based loss measures (Rockafellar and Uryasev 2000; Föllmer and Leukert 2000; Föllmer and Leukert 1999). If the probability-weighted loss remains below a predefined risk budget B , no hedge is indicated; otherwise, the target coverage is

$$\text{Coverage} = \max\left\{0, \frac{\bar{L} - B}{\bar{L}}\right\}.$$

After applying a conservative hedge-efficiency discount and an execution haircut to account for implementation shortfall, the framework outputs an indicative hedge notional rather than a fully executable trading instruction (Almgren and Chriss 2001; Perold 1988).

Bootstrap CVaR at terminal nodes. To complement the point-estimate CAR, a bootstrap CVaR₉₅ is computed at each terminal node. For a given branch path $x^{(s)}$, the fitted model is iterated forward 500 times, adding at each step a residual drawn from out-of-sample residuals constructed via an 80/20 split of the lagged feature matrix $\mathbf{X}_{\text{train}}$: the model is first re-fitted on the first 80% of rows of $\mathbf{X}_{\text{train}}$, and the residuals are taken from the held-out 20% of that matrix. Because $\mathbf{X}_{\text{train}}$ is itself the lag-structured design matrix derived from the

training window, this split operates on the already-lagged observations and ensures that the bootstrap distribution reflects genuinely out-of-sample innovation variance within the training period, rather than in-sample fit to the full lag matrix. When the training window is too short for a meaningful holdout ($n_{\text{lagged}} < 15$ training rows or fewer than 5 holdout rows), the procedure falls back to in-sample residuals. To preserve the autocorrelation structure of the residual series, a block bootstrap with block length $\lfloor \sqrt{n} \rfloor$ is applied (Künsch 1989; Politis and Romano 1994): blocks of consecutive residuals are drawn with replacement and concatenated, so that the temporal dependence within each block is retained. The resulting empirical distribution of terminal cumulative returns is used to compute CVaR_{95} as the mean of the worst 5 % of outcomes, consistent with the expected-shortfall framework of Rockafellar and Uryasev (2000). The hedge-sizing procedure is then applied twice in parallel—once to the point-estimate CAR and once to the CVaR_{95} inputs—and both notionals are reported for comparison. The CVaR-based notional is the more conservative of the two and provides an upper bound on defensible hedge demand given the bootstrap residual distribution.

Rolling CVaR as a time-varying risk diagnostic. To contextualize the forward-looking scenario outputs retrospectively, a 20-day rolling empirical CVaR_{80} of ϵ_t is computed over the historical sample.¹ At each date t the CVaR_{80} is the mean of the worst 20 % of observations in the trailing 20-day window of idiosyncratic returns. Plotting this series alongside the PM return series provides a time-varying picture of left-tail risk, allowing periods of elevated geopolitical risk expectation to be compared visually with periods of heavier downside tails in the electricity residual.

Dual-tail CVaR for long and short exposures. Because the framework is designed to serve both producers (typically long electricity prices) and consumers (effectively short), scenario CVaR is reported for both tails. For a producer or long position, the relevant adverse outcome is a left-tail price collapse: `cvar_long` is the CVaR_{95} of the left tail ($\alpha = 0.95$), stored as `node_cvar_95_long`. For a consumer or short position, the adverse outcome is a right-tail price spike: `cvar_short` is the corresponding right-tail CVaR, stored as `node_cvar_95_short`. Both measures are derived from the same bootstrap distribution of terminal cumulative returns at each scenario node and are reported alongside the point-estimate CAR in the scenario summary table, ensuring that the hedge-sizing output is meaningful regardless of the direction of the underlying exposure. This dual-tail design is consistent with the asymmetric loss treatment described in Rockafellar and Uryasev (2000) and with standard practice in energy-risk management where producers and consumers face structurally opposite tail risks (Hull 2018).

4.1.7 From Model Output to Hedge Design Interpretation

The final step of the methodology is to translate scenario-based hedge demand into an economically interpretable hedging recommendation. The model output indicates whether

¹ CVaR_{95} is not used here because, with a 20-day window, $\lfloor 20 \times 0.05 \rfloor = 1$ tail observation — too few for a reliable tail-mean estimate. CVaR_{80} uses $\lfloor 20 \times 0.20 \rfloor = 4$ observations in the tail, providing a minimum viable tail sample for this window length (Rockafellar and Uryasev 2000).

downside risk is mild, moderate, or severe, and this in turn suggests a corresponding level of hedge coverage or a broad class of instruments consistent with that exposure profile. Different hedging structures—partial option overlays, collars, or more linear protection—involve distinct trade-offs between cost, convexity, and retained upside (Hull 2018; Israelov and Nielsen 2014; Clarke, Silva, and Thorley 2011). The thesis does not attempt to price and compare these structures directly, as doing so would require richer option data, more reliable liquidity information, and detailed execution assumptions. Instead, scenario analysis is used to identify the *kind* of hedge that the magnitude of the modeled downside risk may warrant.

The branching framework is not designed to produce a single “best guess” forecast and then hedge that point estimate. Rather, it stresses the portfolio under adverse but plausible state changes in event expectations and asks how much protection is needed against those states. In that sense, the methodology is closer to a convexity- and fragility-aware risk-management perspective than to a frictionless replication exercise (Taleb 1997). The practical contribution of the framework is therefore not an exact trading recipe, but a disciplined way of linking event-linked predictive models to downside scenario assessment and, from there, to a hedge-sizing signal that can inform broader portfolio decisions.

5 Case Studies

5.1 Overview

To assess the generalizability and practical applicability of the proposed framework, we evaluate it across three case studies. Each case is defined by a clear event window, an associated PM, and a corresponding financial security or market outcome. Taken together, the cases span both short- and longer-horizon settings and include a mix of more traditional and more niche financial exposures. This diversity is useful for evaluating whether the framework remains informative across different market environments, data conditions, and event types.

The case studies also rely on multiple data sources. In principle, all three applications could be adapted to operate with real-time APIs in a production setting, allowing the framework to update forecasts and hedge signals dynamically as new information arrives. For the purposes of this thesis, however, the emphasis is on reproducibility and transparency rather than live deployment. We therefore use static historical datasets downloaded in CSV format wherever possible. In one case study, the `yfinance` package is used to facilitate the retrieval of historical market data in a standardized and reproducible way. All data used in the analysis, together with supporting documentation, are made available through the project’s GitHub repository. Further details on the construction of each dataset and its original source are provided in the relevant case-study sections.

Given that the methodological framework was developed in the previous chapter, the purpose of the present chapter is to examine how it performs in concrete empirical settings. For each case study, we briefly introduce the substantive context, describe how the methodology is adapted to that setting, and summarize the main output at a high level. The goal at this stage is not to provide a full comparative interpretation of results, but rather to establish how the framework behaves when confronted with different forms of event-linked financial data. A more detailed synthesis of findings, including comparisons across cases and discussion of broader implications, is deferred to later sections.

The three case studies were selected according to three criteria: policy salience, data quality, and comparability. Policy salience ensures that the selected events are economically or socially meaningful rather than purely speculative curiosities. Data quality is essential because the framework relies on sufficiently reliable time series for both the financial asset and the related PM. Comparability allows the cases to be studied within a common methodological structure, even when the underlying events and markets differ substantially. Together, these criteria help ensure that the case studies are not only individually interesting, but also informative as a group.

Among the three cases, the Belgian Electricity Prices study is examined in the greatest detail. This case offers the strongest combination of substantive relevance, relatively rich supporting data, and a setting in which the event-linked mechanism is especially interpretable. The remaining two case studies, Argentina and AGI, are retained both to broaden the empirical scope of the thesis and to illustrate the boundaries of the framework. In that sense, they serve not only as additional applications, but also as stress tests that reveal where the method performs less cleanly or where the available data impose meaningful limitations. The choice to emphasize the Belgian Electricity Prices case more heavily was itself guided

by the same three selection criteria used to construct the broader case-study set.

Table 1: Overview of case-study selection

Case study	Short description	Policy salience	Data quality	Comparability
Belgian Electricity Prices	Power-market case linking asset returns to nuclear-detonation event expectations in a lower-frequency (daily) setting.	High: energy security and conflict are politically and economically salient, and the case links directly to recent debates over banning such markets.	Strong: market and contextual data are sufficiently rich for modeling and scenario analysis, although the opacity of energy derivatives markets limits the analysis somewhat.	High: well suited for comparison with other event-driven asset cases.
Argentina	Country-risk case examining how event expectations related to Argentina map into financial-market responses.	High: political developments in Argentina are clearly relevant for investors and broader macro-financial risk.	Moderate: financial and PM data allow application of the framework, although the lack of higher-frequency data limits statistical power.	High: provides a sovereign and macro-political contrast to the other cases.
AGI	Technology-focused case studying market responses to event expectations related to artificial general intelligence.	High: AGI is increasingly relevant for regulation, industrial policy, and strategic competition.	Moderate to weak: the signal is more novel and potentially noisier, but still adequate for exploratory application.	Moderate: extends the framework to a less traditional but still policy-relevant domain.

5.2 Belgian Electricity Prices

5.2.1 Background and Goals

European electricity prices are formed in a market that is highly integrated, but not frictionless. Through the Single Day-Ahead Coupling (SDAC) mechanism, wholesale electricity is traded across Europe in a common auction framework, yet prices remain zonal and can diverge substantially when interconnector capacity is constrained or when national generation mixes differ (ENTSO-E 2026c; Gasparella, Koolen, and Zucker 2023; Garcia et al.

2018). In practice, day-ahead electricity prices reflect the interaction of demand, fuel costs, carbon prices, renewable intermittency, storage availability, and transmission constraints. This broader European structure is important for the present case study because it implies that geopolitical shocks need not affect Belgian electricity prices only through domestic production conditions; they can also propagate through cross-border market integration and regional fuel-price transmission (Cevik and Zhao 2025; Gasparella, Koolen, and Zucker 2023).

The most important recent shock motivating this case is Russia’s full-scale invasion of Ukraine and the associated European gas-supply crisis. The collapse in Russian pipeline gas deliveries contributed to a sharp rise in European gas and electricity prices, while gas-fired plants retained a disproportionate role in setting wholesale electricity prices under the merit-order system (International Energy Agency 2025; Gasparella, Koolen, and Zucker 2023). IMF evidence further suggests that volatility spillovers across European electricity markets intensified after the invasion, reinforcing the view that geopolitical disturbances can transmit across borders rather than remaining confined to individual national markets (Cevik and Zhao 2025). The conflict in the Middle East may also be understood as an additional source of energy-market volatility, but the strongest and best-documented structural driver of recent European electricity-price dislocation remains the Ukraine-induced gas shock.

Belgium is an especially informative case within this broader setting. Belgian wholesale electricity prices reflect a generation mix that includes substantial nuclear capacity, material gas exposure, a growing renewable share, and deep interconnection with neighboring systems. According to the Belgian Federal Public Service Economy, Belgium’s electricity system in 2024 combined a large nuclear component with rising renewable generation and continued net electricity imports, making Belgian prices a useful aggregate outcome for studying how geopolitical risk propagates through both domestic generation and regional market channels (Belgian Federal Public Service Economy 2025). This interpretation is consistent with earlier work showing that Belgian day-ahead electricity prices are materially shaped by market integration, especially with France (Garcia et al. 2018). For the purposes of this thesis, electricity prices are therefore more informative than a single fuel price, because they embed the combined effect of generation mix, interconnection dependence, backup thermal generation, and neighboring-market conditions.

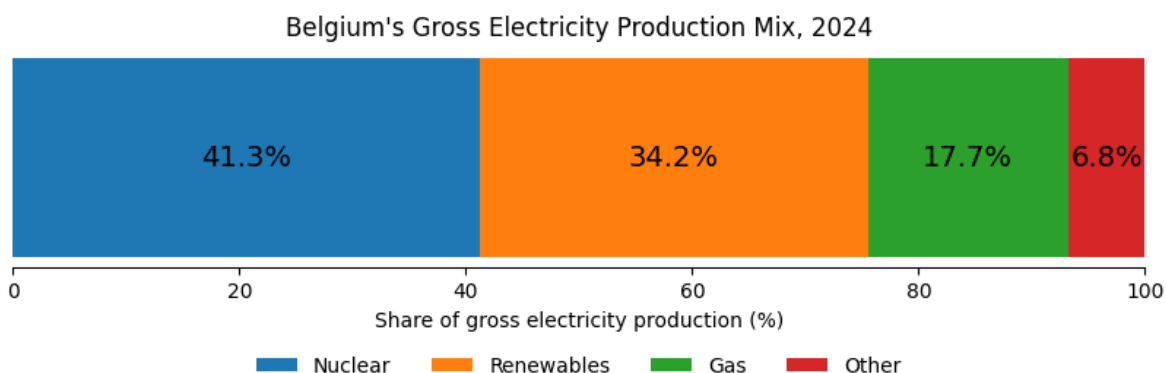


Figure 1: Source: Belgian Federal Public Service Economy, *Belgian Energy Data Overview* (June 2025), reproduced by the author.

Belgium is also relevant from an energy-security perspective. The IEA describes Belgium as a highly interconnected gas hub with 17 cross-border gas interconnection points and the Zeebrugge LNG terminal, which serves both domestic demand and the wider north-west European market (International Energy Agency 2022). Since the invasion of Ukraine, Belgium has diversified its natural-gas sourcing more strongly toward LNG, while the wider energy crisis contributed to a political reversal of the planned nuclear phase-out through the extension of Doel 4 and Tihange 3 (International Energy Agency 2022; Belgian Federal Public Service Economy 2025). The nuclear-fuel channel should nevertheless be described with some care. The war heightened broader European nuclear-fuel security concerns, particularly around conversion and enrichment services, and later fuel contracts for the Belgian reactor extension excluded Russian-origin materials (Synatom 2021; Synatom 2024; Gasparella, Koolen, and Zucker 2023). This makes the Belgian electricity market a plausible setting in which changes in perceived geopolitical tail risk may affect prices through gas markets, interconnection channels, and energy-policy expectations rather than through a single, literal supply mechanism.

To study this relationship, we use daily electricity price data from the ENTSO-E Transparency Platform, which provides publicly available European electricity market data, and from Ember, which republishes cleaned and aggregated European wholesale electricity price series sourced primarily from ENTSO-E (Hirth, Mühlenpfordt, and Bulkeley 2018; ENTSO-E 2026a; Ember 2025). We focus on the Belgian day-ahead market to narrow the empirical scope while preserving the broader European transmission channels discussed above. These prices are then compared with daily odds from the prediction market “*Nuclear weapon detonation in 2025?*” The contract is not interpreted as a literal forecast of Belgium-specific physical disruption; rather, it is treated as a proxy for shifts in perceived geopolitical tail risk. It is important to be explicit about the limits of this causal chain. A nuclear detonation would be a catastrophic geopolitical event, but the specific mechanism through which it would affect Belgian day-ahead electricity prices in the same direction and at the same daily frequency as the ongoing Russia–Ukraine gas supply shock is not obvious. The thesis does not claim a structural channel from nuclear-detonation probability to Belgian electricity prices; instead, the PM is used as a sentiment and risk-appetite indicator that may co-move with broader geopolitical risk perceptions affecting energy markets. The empirical findings — including the 44% PM importance share under Ridge and the non-zero OOS predictive content — are interpreted as consistent with this proxy interpretation, not as evidence of a direct causal link. Replication across multiple years would be necessary before concluding that a structural relationship exists. Historical market prices are retrieved from Polymarket’s price-history interface and related market documentation (Polymarket 2026; Polymarket 2025). In this way, the case study examines whether movements in a conflict-related PM contain information about subsequent changes in Belgian wholesale electricity prices.

5.2.2 Variables and Data Cleaning

As noted above, the electricity-price data are obtained from ENTSO-E in CSV format (ENTSO-E 2026a). In the notebook implementation, we use Belgian *day-ahead* wholesale electricity prices for 2025 rather than actual intraday settlement prices. The raw file con-

tains the columns **MTU** (CET/CEST), which records the market time unit in local Belgian time, **Area**, which is used to isolate the Belgian bidding zone, and **Day-ahead Price** (EUR/MWh). After filtering to Belgium, the timestamps are converted from Europe/Brussels time to UTC, and the price column is recoded as **Price** (EUR/MWh) for consistency with the downstream analysis. The resulting electricity series is quarter-hourly, with up to 96 observations per day, which makes it substantially higher-frequency than the prediction market data used in the final model.

To align the energy data with the prediction market series, the notebook aggregates Belgian electricity prices to daily frequency. Specifically, quarter-hourly Belgian prices are averaged within each day to form a daily series. The code also retains only days with complete intraday coverage and builds a balanced daily panel before extracting the Belgian series, which appears in the merged dataset under the column **Country Avg - Belgium**. This cleaning step ensures that the daily electricity series is not distorted by missing intraday intervals. The focus on daily data is ultimately driven by the frequency of the prediction market data available for this thesis.

Figure 2 shows the resulting daily Belgian electricity-price series. The series is clearly volatile, with elevated prices in the early part of the year, several short-lived spikes, and occasional very low or even negative values. This reinforces the need to model electricity prices as a noisy market outcome shaped by both domestic conditions and broader regional market pressures.

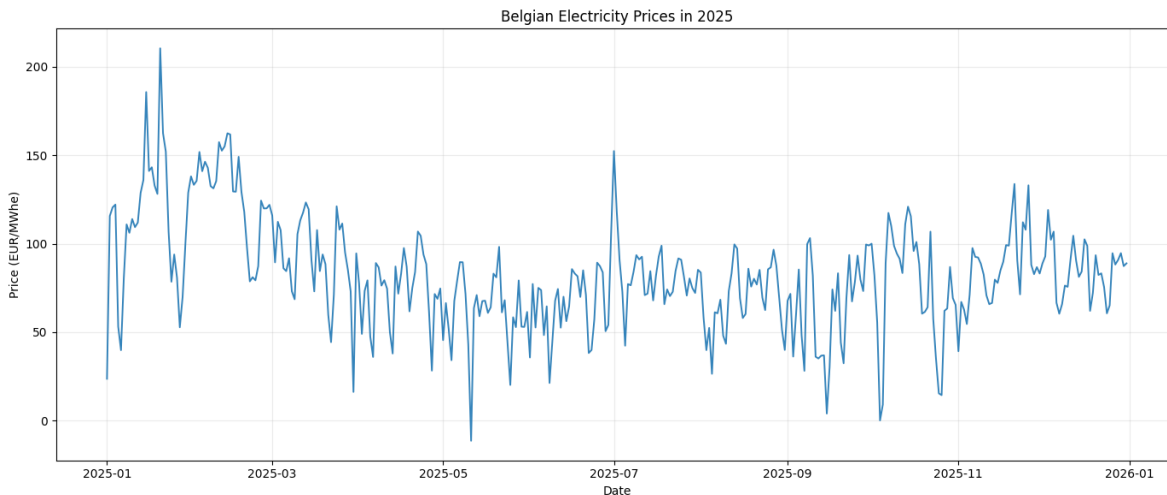


Figure 2: Daily Belgian day-ahead electricity prices in 2025, aggregated from quarter-hourly ENTSO-E observations. Source: ENTSO-E Transparency Platform, processed by the author.

Prediction market data are downloaded in CSV format from Polymarket Analytics using the contract “*Nuclear weapon detonation in 2025?*” (Polymarket Analytics 2026). This contract resolves on 2026-01-01, paying \$1 if a nuclear weapon is detonated in 2025 and \$0 otherwise; the time-to-expiry variable used throughout the modeling therefore counts down from 365 days at the start of the sample to 0 at the resolution date. In the notebook, the raw CSV contains a date-like column labeled **category** and a price column labeled **Yes**. These are

converted into a standardized daily panel with the variables `Date` and `price.Yes.nuclear`. A backward-compatible alias `price.Yes` is then created, together with the complementary `price.No = 1 - price.Yes`. Because the raw daily market series contains some gaps, the notebook fills missing values by linear interpolation and then aligns the completed prediction market series to a full daily 2025 calendar. This introduces an important limitation: the final analysis is conducted at daily frequency, even though the underlying electricity market operates at much higher temporal resolution. A higher-frequency prediction market history would be preferable, but obtaining such data was beyond the budget of the thesis.

Figure 3 displays the cleaned daily YES and NO prices for the nuclear-detonation market. The YES price trends downward over the course of the year, while the NO price correspondingly approaches one. Substantively, this suggests that the market assigned a declining probability to a nuclear detonation as 2025 progressed. From a modeling perspective, the figure also makes clear that the prediction market series evolves much more smoothly than the electricity-price series, which limits the amount of short-run variation available for identification at daily frequency.

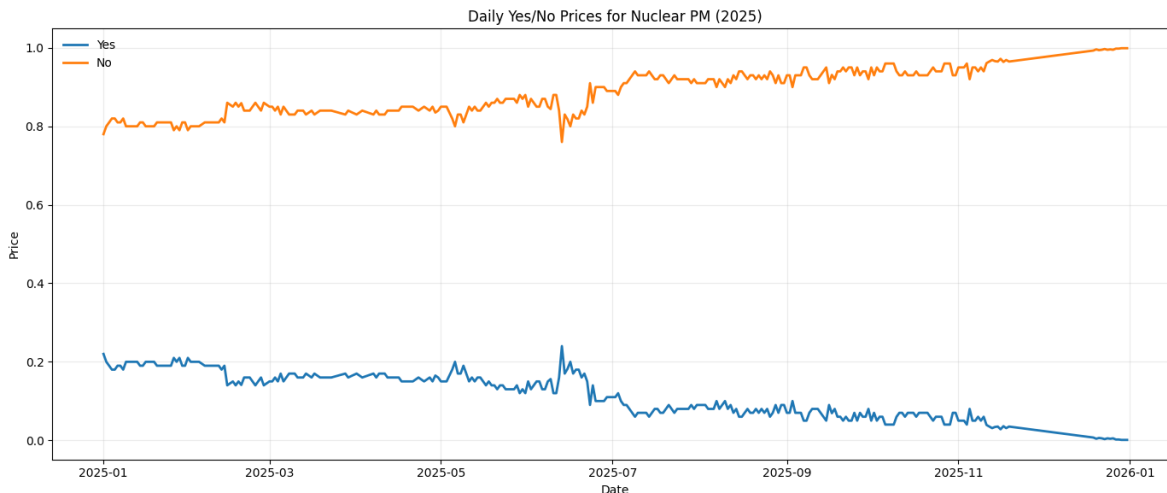


Figure 3: Daily YES and NO prices for the Polymarket contract “Nuclear weapon detonation in 2025?” Source: Polymarket Analytics, processed by the author.

In addition to these two central datasets, the notebook constructs a set of explanatory variables used later in the factor model. Belgian load data are imported from ENTSO-E power statistics in CSV format (ENTSO-E 2026b). The key load columns are **Actual Total Load (MW)** and **Day-ahead Total Load Forecast (MW)**, which are first converted to UTC and then averaged to daily frequency. From these, the notebook derives daily percentage-change variables for actual load, forecast load, and forecast error. Weather data are obtained from Open-Meteo using the `openmeteo-requests` client (Open-Meteo 2026; Patrick Zippenfenig 2026). Using Brussels as a proxy for Belgium, the notebook retrieves daily temperature, precipitation, and wind-speed measures for 2025 and merges them into the daily electricity-price panel. The final aligned dataset is therefore daily throughout and combines Belgian electricity prices, the nuclear-detonation PM series, load variables, and weather variables in a common UTC-based time frame.

5.2.3 The Factor Model

Before modeling the dynamic relationship between Belgian electricity prices and the prediction market series, we first isolate the component of electricity-price returns that is plausibly attributable to broad weather and demand conditions. The purpose of this step is not to build a complete structural model of wholesale electricity formation, but rather to remove a parsimonious block of systematic variation so that the remaining series can be interpreted as an idiosyncratic return and passed into the downstream time-series model. Formally, we write daily Belgian electricity returns as

$$r_t = g(f_t) + \epsilon_t,$$

where f_t denotes the vector of observed weather- and load-related factors, $g(\cdot)$ is the estimated factor mapping, and ϵ_t is the residual component that is carried forward as the idiosyncratic return.

The factor choice is well grounded in the electricity-price literature. In wholesale power markets, prices are determined on a steep merit-order supply stack, so realized demand pressure, weather-driven stress, and unexpected system imbalance are first-order drivers of price variation. Daily temperature is therefore a natural factor because it shifts both heating and cooling demand and can also affect operational constraints; actual load captures the level of demand pressure in the system; and the gap between realized and forecast load captures unexpected system stress that was not fully priced into the day-ahead market at gate closure (Huisman 2008; Mosquera López, Uribe, and Joaqui-Barandica 2024; Forbes and Zampelli 2014; Jónsson et al. 2013; Möbius et al. 2025; Eicke, Ruhnau, and Hirth 2021; ENTSO-E 2025). In the notebook implementation, this core factor block is augmented with additional weather controls, namely precipitation and wind speed, so that the residualization step does not attribute broad meteorological variation to event-specific price movements.

To assess whether the load factors contribute meaningfully beyond the weather block alone, we estimate two versions of the PCA factor model. Following the methodology described above, the SVD is fitted on a held-back training window of $n_{\text{train}} = \max(30, n - 20)$ observations — with the 363-observation daily sample this gives 343 training observations — and the fixed loadings are then applied to the full series. We compare a weather-only and a weather-plus-load specification under this training-window-only estimation scheme. The weather-only specification retains three active factors and three principal components, yielding an R^2 of 0.011 and an adjusted R^2 of 0.003 with a residual standard deviation of 557.69. Adding the three load-based factors — percentage changes in actual load, forecast load, and the forecast-versus-actual gap — expands the active factor set to six and increases the retained components to five, raising R^2 marginally to 0.011 and reducing the residual standard deviation to 557.68. The adjusted R^2 turns slightly negative (-0.003) under the expanded specification, reflecting the penalty for the additional parameters relative to the modest incremental fit. These results indicate that the weather and load factors together explain only a small share of day-to-day variation in the symmetric Belgian return, which is unsurprising at daily frequency where broader market forces dominate. Nevertheless, including the load block is retained in the final specification because it removes a defensible additional source of systematic variation — particularly the forecast-error channel — even when its marginal R^2 contribution is small. The more important justification is theoretical: omitting load factors

would risk attributing demand-shock-driven residuals to the event-linked prediction market signal in the downstream model.

Methodologically, the model is designed to be stable under the collinearity and outlier structure typical of electricity data. The raw factor matrix is first restricted to non-constant series and standardized. Because weather and load variables are strongly correlated, the standardized factor block is then orthogonalized using principal components analysis, and the number of retained components is chosen by a cumulative explained-variance rule that preserves 95% of total factor variation. Table 2 reports the component-count sensitivity for the full six-factor specification. The R^2 and RMSE are broadly flat from $k = 1$ through $k = 4$, with a more meaningful improvement at $k = 5$, which is where the cumulative explained variance crosses 95% (0.987). Adding a sixth component produces only marginal further improvement ($R^2 = 0.014$, RMSE = 556.14), confirming that the 95% threshold is a reasonable and stable stopping rule.

Table 2: Component-count sensitivity for the PCA factor model. R^2 , RMSE, and residual standard deviation are reported for $k = 1$ to $k = 6$ retained principal components. The 95% cumulative explained-variance threshold is crossed at $k = 5$, and results are broadly stable beyond that point. Source: author’s calculations.

k	R^2	RMSE	Residual std
1	0.000	559.98	560.76
2	0.001	559.61	560.39
3	0.001	559.61	560.38
4	0.002	559.60	560.37
5	0.011	556.91	557.68
6	0.014	556.14	556.91

This dimensionality reduction helps stabilize estimation and avoids attributing undue importance to any one collinear regressor, as is visible in the PCA loadings: PC1 is dominated by the two load variables (loadings of -0.705 and -0.702 for actual and forecast load respectively), while PC2 captures the precipitation–wind speed axis (loadings of 0.687 and 0.686), and PC3 isolates the forecast-error and temperature channels. A baseline linear factor mapping is first estimated on the retained principal-component scores. To make the final residualization more robust, however, the notebook replaces plain percentage returns with a symmetric return transform and estimates the factor mapping using a Huber robust regression on the retained principal components (Jolliffe 2002; Huber 1964). As shown in Figure 4, five principal components are sufficient to exceed the 95% cumulative explained-variance threshold.

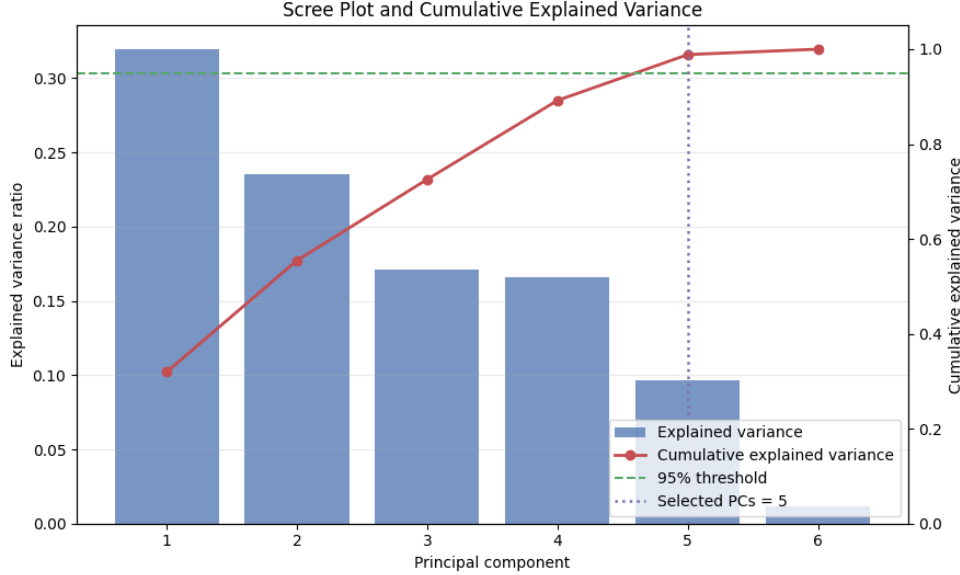


Figure 4: Scree plot and cumulative explained variance for the standardized weather- and load-based factor block. The first five principal components explain at least 95% of total factor variance and are therefore retained in the final factor model. Source: author’s calculations.

The resulting specification is statistically more defensible than a simple raw-factor OLS regression. First, it addresses multicollinearity through PCA. Second, it reduces the influence of outlying observations through robust estimation rather than allowing a few price spikes to dominate the fitted coefficients. Third, it keeps the factor model deliberately parsimonious: the goal here is not to fully explain Belgian wholesale electricity prices, but to remove the most defensible weather- and demand-related common component before the later predictive analysis. This is consistent with the broader forecasting literature, which shows that demand, weather, forecast information, and related system fundamentals materially improve electricity-price models, even when richer specifications with fuels, renewables, and cross-border flows are available (Billé et al. 2023; Jónsson et al. 2013; Möbius et al. 2025).

The notebook implements a substantial validation battery for this residualization step, and the key numerical results are reported here. Serial dependence in the residuals is assessed using the Ljung–Box test at lags 5, 10, and 20, and the Durbin–Watson statistic. The Ljung–Box statistics are 1.53 ($p = 0.909$), 1.64 ($p = 0.998$), and 2.50 ($p \approx 1.000$) at lags 5, 10, and 20 respectively, providing no evidence of remaining autocorrelation at any horizon. The Durbin–Watson statistic of 1.872 is consistent with the absence of first-order serial correlation. Heteroskedasticity is examined using the Breusch–Pagan, White, and ARCH tests. None of the three tests reject at conventional levels: the Breusch–Pagan statistic is 5.02 ($p = 0.414$), the White statistic is 16.73 ($p = 0.671$), and the ARCH LM statistic at lag 5 is 0.017 ($p \approx 1.000$). The absence of detectable ARCH effects is notable given that electricity prices are known to exhibit volatility clustering at intraday frequencies; at daily frequency, after residualization, this structure appears to be substantially attenuated. Taken together, these diagnostics indicate that the robust factor model residuals do not exhibit systematic

serial dependence or conditional heteroskedasticity that would invalidate their use as inputs to the downstream time-series models.

Influence diagnostics identify one observation with a substantially elevated Cook’s distance: 5 October 2025 (Cook’s $D = 0.972$), with a studentized residual of 136.08, indicating an extreme price spike on that date that the factor model does not fully absorb. The next most influential observations — 6 October 2025 ($D = 0.007$), 2 January 2025 ($D = 0.006$), and 12 May 2025 ($D = 0.006$) — have Cook’s distances two orders of magnitude smaller. The dominance of the 5 October observation is a direct consequence of its extreme studentized residual and is the primary motivation for using Huber robust regression rather than OLS in the final specification: the robust estimator substantially down-weights this observation, preventing it from distorting the fitted factor coefficients for the remainder of the sample.

Figure 5 summarizes the most important visual diagnostics for the final robust factor model. The residuals remain heavy-tailed, which is unsurprising in electricity-price data, but there is no strong residual trend in the fitted plot and the autocorrelation profile is substantially muted relative to the raw series. This is sufficient for the residualization step to serve its intended purpose as a preprocessing device.

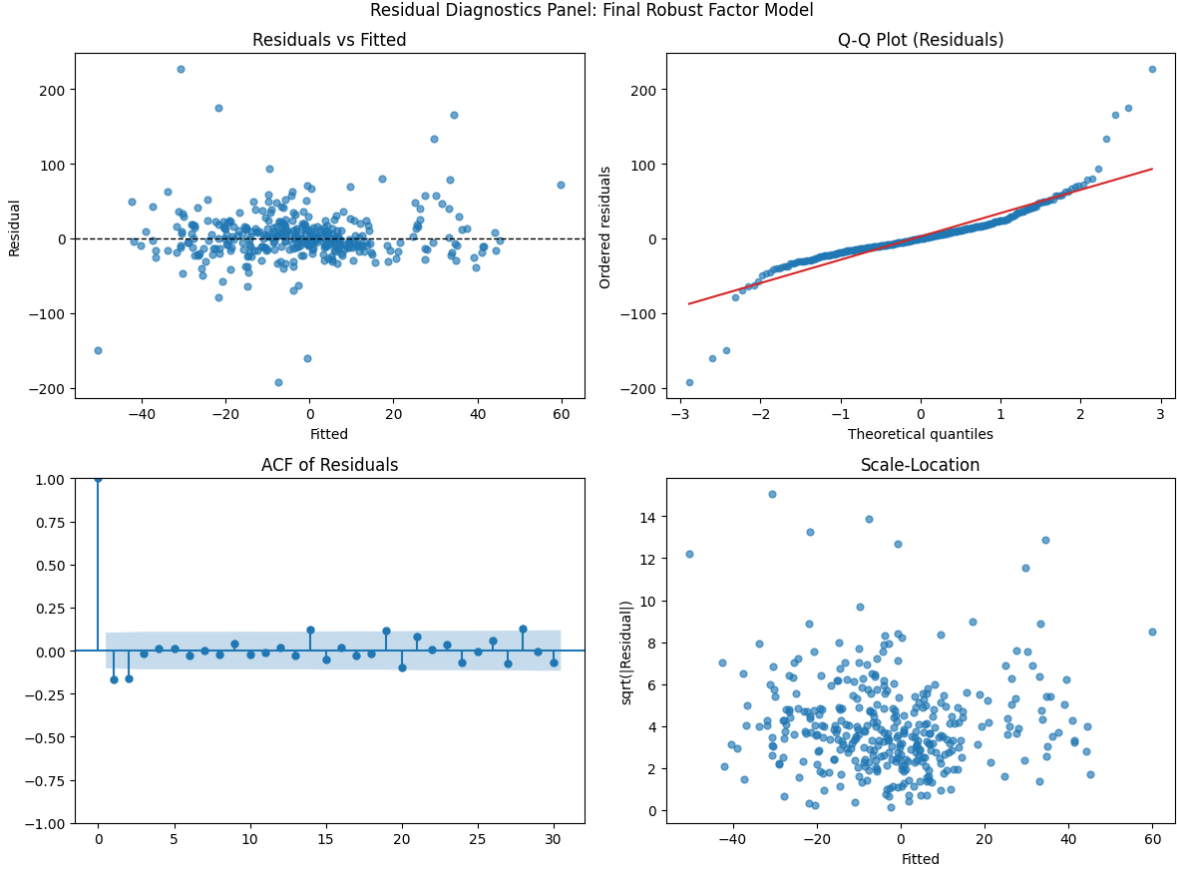


Figure 5: Residual diagnostics for the final robust factor model. The panel reports residuals versus fitted values, a Q–Q plot, the residual autocorrelation function, and a scale-location plot. Together, these diagnostics are used to assess whether the residualization step leaves strong systematic misspecification, serial dependence, or unstable variance before the residual series is carried forward into the downstream time-series model. Source: author’s calculations.

The interpretation of the residual should nevertheless remain modest. The factor model assumes that a meaningful portion of daily electricity returns can be decomposed additively into a systematic weather-demand component and a residual component, conditional on the chosen information set. Because the analysis is conducted at daily frequency, some intraday scarcity dynamics and spike behavior are necessarily smoothed out. In addition, omitted variables such as fuel prices, cross-border flows, outages, or renewable-generation forecasts may still contribute to the residual. The dominance of the 5 October spike in the influence diagnostics is a concrete illustration of this limitation: that observation likely reflects an event-specific market stress that the weather-and-load factor block was never designed to capture. Accordingly, ϵ_t should be interpreted as idiosyncratic only relative to the included daily weather- and load-based factor block, rather than as a fully structural shock. This is sufficient for the present thesis, because the residual is used as a cleaned return series for subsequent time-series modeling rather than as the object of standalone economic interpretation.

As its final output, the robust factor model produces three series: the transformed Belgian return, the fitted systematic component, and the residual component `BEL_idiosyncratic`. It is this last series that is carried forward into the predictive modeling stage. In this way, the factor model serves as a statistically disciplined preprocessing step that removes a defensible common weather-demand signal while preserving a residual series more suitable for event-linked time-series analysis.

5.2.4 Time Series Data Exploration

Having isolated the residual component from the factor model, we next examine the time-series structure of the two central variables used in the predictive stage: the Belgian idiosyncratic electricity return series, denoted `BEL_idiosyncratic`, and the return series derived from the nuclear prediction market, denoted `PM_nuclear_returns`. The purpose of this step is exploratory. Before fitting benchmark VAR and machine-learning models, it is useful to inspect the marginal distributions, temporal variation, short-run dependence structure, and stationarity of each series. In particular, because the factor model residual becomes the variable of interest on the electricity side, any remaining autocorrelation in `BEL_idiosyncratic` is precisely the dependence structure that the later time-series models are designed to capture.

Figure 6 shows the exploratory plots for the Belgian idiosyncratic return series. The time-series panel indicates that, although much of the broad weather- and demand-related component has been removed, the residual series still exhibits substantial volatility and several large spikes. The histogram confirms that the distribution is centered near zero but clearly heavy-tailed, which is typical of electricity-price returns even after residualization. The ACF and PACF panels in Figure 7 suggest that most higher-order dependence has been removed by the factor model, but some short-run serial structure remains, particularly at the first few lags. This is an encouraging result for the next stage: the residual series appears sufficiently cleaned to move beyond simple factor regression, while still retaining dynamic structure worth modeling.

To verify that the residualized series is suitable for standard time-series modeling, we apply two complementary formal stationarity tests. The Augmented Dickey–Fuller (ADF) test, which takes a unit root as its null hypothesis, strongly rejects non-stationarity for `BEL_idiosyncratic` ($p < 0.01$), consistent with the visual impression of a mean-reverting process. The KPSS test, which reverses the null by treating stationarity as the maintained hypothesis, fails to reject at the 5% level, corroborating the ADF conclusion. Taken together, these results support treating the Belgian idiosyncratic return series as stationary in the downstream models, which is a necessary condition for the validity of the VAR benchmark.

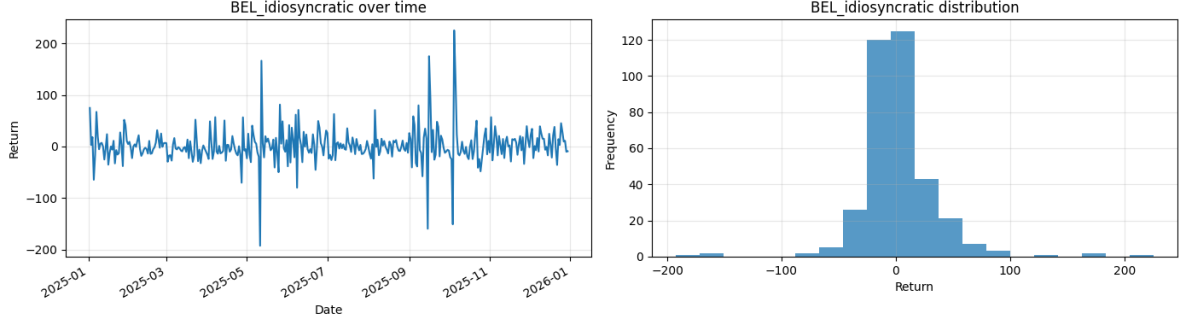


Figure 6: Exploratory time-series plots for `BEL_idiosyncratic`, including the series over time and its empirical distribution. The factor-model residual is centered around zero but remains volatile and heavy-tailed. Source: author’s calculations.

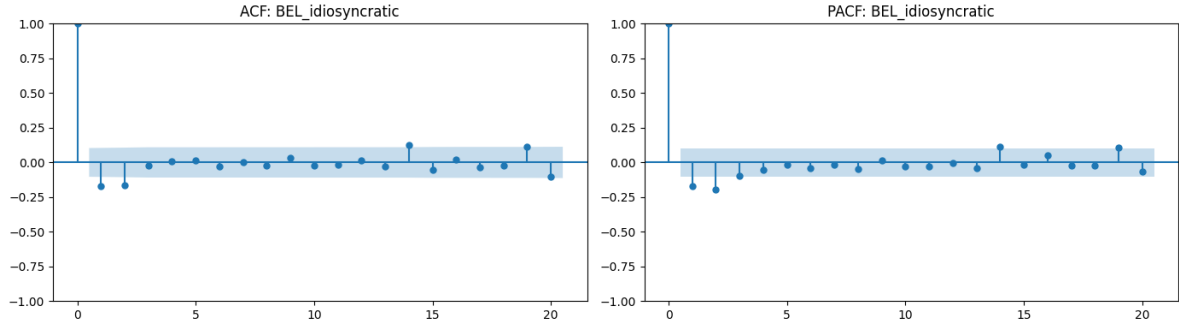


Figure 7: Autocorrelation and partial autocorrelation functions for `BEL_idiosyncratic`. The factor model substantially reduces systematic structure, but limited short-run dependence remains, motivating the subsequent time-series modeling stage. Source: author’s calculations.

We then examine the nuclear PM return series. Figure 8 shows that `PM_nuclear_returns` is also centered near zero, but its behavior differs materially from the Belgian residual series. The return process is much quieter early in the sample and becomes more volatile later in the year, with several sharp positive and negative jumps. The histogram again suggests a peaked center and non-negligible tails. The associated ACF and PACF plots in Figure 9 indicate limited but non-zero short-run dependence, especially at the first lag. Relative to the electricity residuals, the prediction market series appears more episodic, with bursts of activity rather than persistent day-to-day oscillation. This difference in behavior is important because it suggests that the joint system is unlikely to be well described by a purely smooth linear dependence structure alone.

Formal stationarity tests for `PM_nuclear_returns` yield a more nuanced picture. The ADF test rejects the unit-root null at the 5% level, suggesting that the series does not contain a stochastic trend. However, the KPSS test also rejects its stationarity null at the same significance level, pointing to some form of structural instability in the level or variance of the series over the year. This tension is unsurprising given the visual evidence: the prediction market series is not trended in the classical sense, but it exhibits a clear

shift in volatility regime in the second half of the sample, which the KPSS test is sensitive to. For practical purposes, working with the percentage-change transformation of the YES price—rather than the level—addresses the most salient source of non-stationarity, and the ADF result provides sufficient justification for proceeding with standard lag-based modeling. The volatility shift is, however, a reminder that the relationship between this series and the Belgian residual may not be fully stable across the year, a consideration that informs the use of rolling and expanding validation windows in the subsequent model-comparison stage.

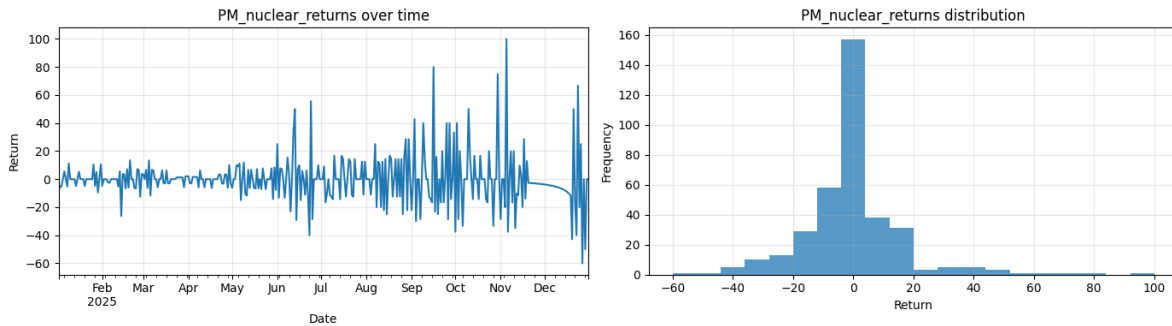


Figure 8: Exploratory time-series plots for `PM_nuclear_returns`, including the series over time and its empirical distribution. The series is centered near zero but becomes more volatile later in the sample and exhibits occasional large jumps. Source: author’s calculations.

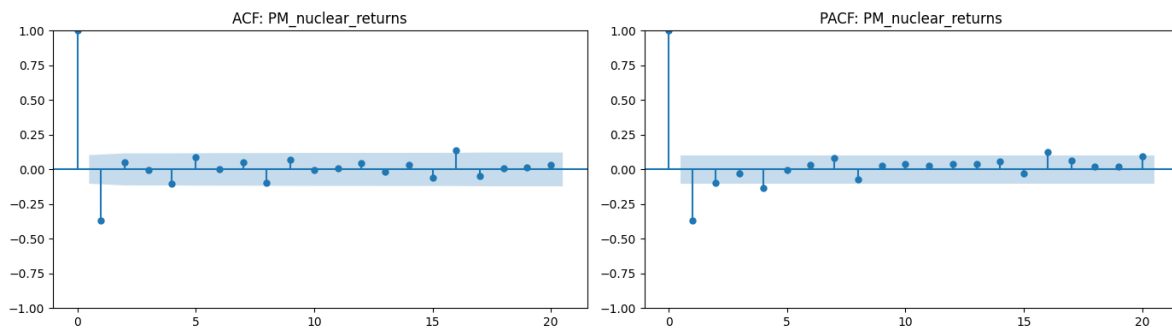


Figure 9: Autocorrelation and partial autocorrelation functions for `PM_nuclear_returns`. The series displays limited short-run dependence, particularly at early lags, which informs the lag structure considered in the initial benchmark models. Source: author’s calculations.

Taken together, these exploratory plots and stationarity diagnostics motivate two aspects of the modeling strategy described below. First, the residualized Belgian return series is stationary and exhibits enough short-run autocorrelation to justify explicit time-series modeling rather than treating observations as conditionally independent. Second, the nuclear PM return series, while not trended, shows signs of volatility non-stationarity that make it an episodic and bursty signal; benchmark models should therefore be flexible enough to allow for weak persistence, asymmetric responses, and occasional nonlinear behavior. These observations do not determine the final model, but they provide a useful empirical basis for the initial benchmark specifications, for the lag ranges considered in subsequent model selection, and for the emphasis on time-respecting rather than i.i.d. cross-validation throughout.

5.2.5 Benchmark and Model Comparison

We next compare a set of benchmark time-series models to determine whether the residualized Belgian return series is better described by a simple linear dynamic structure or by a more flexible predictive model. This comparison serves two purposes. First, it provides an empirical check on the exploratory findings from the previous subsection: the remaining dependence in `BEL_idiosyncratic` is modest, noisy, and potentially nonlinear. Second, it determines which model class is most appropriate for the scenario simulation that follows.

We begin with a broad model bakeoff on the daily series `PM_nuclear_returns` and `BEL_idiosyncratic`. This first task compares traditional benchmark models, a GARCH specification, regularized linear machine-learning models, kernel methods, and tree-based methods under a common holdout-based forecasting exercise. Table 3 reports the resulting ranking. Two findings stand out immediately. First, the regularized linear and kernel methods outperform the tree-based ensemble models in the bakeoff, with `SVR` (rank 1), `SVR_linear` (rank 2), and `Ridge` (rank 3) at the top. This suggests that at bakeoff frequency the residualized Belgian electricity return is better described by regularized linear or kernel-based relationships rather than flexible tree ensembles. Second, the traditional VAR benchmark performs respectably at rank 6, ahead of GARCH and the tree-based models, reflecting the relatively modest nonlinear structure in this dataset.

Table 3: Model bakeoff ranking for the Belgian electricity case (Task 1)

Model	Family	Best RMSE	Median RMSE	Best MAE	Rank
<code>SVR</code>	ML	22.097	22.097	18.195	1
<code>SVR_linear</code>	ML	22.155	22.155	18.248	2
<code>Ridge</code>	ML	22.876	22.876	18.340	3
<code>Lasso</code>	ML	22.880	22.880	18.343	4
<code>ElasticNet</code>	ML	22.888	22.888	18.351	5
<code>VAR</code>	Traditional	23.036	23.036	18.519	6
<code>GARCH</code>	GARCH	23.180	23.180	18.585	7
<code>FoundationModel</code>	Foundation	23.780	23.780	18.923	8
<code>ExtraTrees</code>	ML	23.862	23.862	18.903	9
<code>GradientBoosting</code>	ML	24.672	24.672	19.306	10
<code>RandomForest</code>	ML	26.331	26.331	21.195	11

To avoid selecting a model on the basis of a single bakeoff alone, the notebook then applies the chronology-preserving validation framework described earlier in the thesis. In this second task, shortlisted models are evaluated under both expanding-window and rolling-window procedures, with forecast accuracy and directional accuracy computed across repeated forward-looking splits. Table 4 reports the resulting generalizability ranking. Here the picture consolidates further toward the regularized linear class. The best out-of-sample generalization is achieved by `Ridge` (validation rank 1), closely followed by `ElasticNet` and `Lasso`. `ExtraTrees` retains a competitive expanding-window RMSE but underperforms on the rolling window, while `GradientBoosting` and the tuned `FoundationModel` generalize substantially less well than they performed in the initial bakeoff, a pattern consistent with overfitting to local nonlinear structure that does not persist across temporal splits.

Table 4: Generalizability ranking under expanding and rolling validation (Task 2)

Model	Mean RMSE (exp.)	Mean RMSE (roll.)	Dir. acc. (exp.)	Dir. acc. (roll.)
Ridge	37.248	37.766	0.579	0.543
ElasticNet	37.248	37.786	0.579	0.543
Lasso	37.252	37.785	0.579	0.543
ExtraTrees	37.023	38.426	0.557	0.543
RandomForest	37.822	38.940	0.529	0.550
SVR	37.708	37.795	0.529	0.500
SVR_linear	38.566	38.704	0.507	0.536
FoundationModel	43.007	46.919	0.571	0.529
GradientBoosting	48.440	51.374	0.500	0.529

These two tables are jointly revealing. Both the bakeoff and the generalizability task favor regularized linear models, with SVR winning the bakeoff and Ridge winning the validation. This convergence indicates that the predictive signal linking Belgian idiosyncratic electricity returns to lagged nuclear prediction market returns is, on the available data, better characterized by a stable linear mapping than by flexible nonlinear ensemble methods. Tree-based models such as ExtraTrees, RandomForest, and GradientBoosting appear to overfit the bakeoff sample and generalize less reliably. This is a meaningful finding: it suggests that whatever predictive content the nuclear prediction market holds for Belgian electricity residuals is modest and approximately linear in nature.

To reconcile these two tasks, the notebook forms a combined ranking across the bakeoff and validation stages. Table 5 summarizes the resulting decision. SVR wins the initial bakeoff, and Ridge wins the validation task; the final simulation model is **Ridge**. This reflects its strong and stable performance across both tasks: it ranks third in the broad bakeoff (only marginally behind SVR and SVR_linear) while also winning the generalizability exercise outright. The linear structure imposed by Ridge is not a concession but an empirical finding: the data support it.

Table 5: Final selection summary across both model-comparison tasks

Model	Bakeoff rank	Best RMSE	Best MAE	Family	Final role
Ridge	3	22.876	18.340	ML	Final simulation model
SVR	1	22.097	18.195	ML	Bakeoff winner
ElasticNet	5	22.888	18.351	ML	Validation runner-up
Lasso	4	22.880	18.343	ML	Validation runner-up
ExtraTrees	9	23.862	18.903	ML	Validation rank 4

The substantive implication is that the Belgian case is governed by a relatively stable, approximately linear predictive structure — the opposite of what might be expected if prediction market shocks triggered nonlinear threshold responses in electricity prices. The dominance of Ridge and its close relatives (Lasso, ElasticNet) across both tasks is consistent with the exploratory finding that both the electricity residual series and the nuclear PM

return series are only modestly persistent and do not exhibit strong interaction patterns. For this reason, **Ridge** is retained as the final simulation model, which provides a natural segue to the next subsection.

5.2.6 Final Model Specification

The final predictive specification retained for this case study is a **Ridge** regression model. This choice is based on its balanced performance across both model-comparison tasks: it is the bakeoff rank-3 model and the outright validation winner, and the combination of these two rankings produces the lowest combined rank score in the final selection. In the final simulation setup, the selected model uses three lags of the target and the prediction market driver, together with the time-to-expiry feature, giving seven predictors in total. The fitted pipeline reports an in-sample R^2 of approximately 0.114 and an out-of-sample R^2 of -0.079 from the backtest, indicating that while the model captures some in-sample structure, its predictive power over held-out periods is limited (Hoerl and Kennard 1970; Bergmeir and Benítez 2012; Hyndman and Athanasopoulos 2021).

Substantively, the selection of **Ridge** confirms that the remaining predictive signal in `BEL_idiosyncratic` is modest and approximately linear. The advantage of regularized linear models over tree-based ensembles in the generalizability task indicates that whatever linkage exists between Belgian idiosyncratic electricity returns and lagged nuclear prediction market returns is not strongly nonlinear or state-dependent on the available data. At the same time, the feature-importance output, computed using absolute coefficient magnitudes evaluated on the held-out 20% of the training window, reveals that the prediction market lags contribute meaningfully alongside the autoregressive terms.

Table 6: Normalized feature importance shares for the selected **Ridge** model. Raw absolute coefficient magnitudes are normalized to sum to 1.0 by dividing by their total. Source: author’s calculations.

Feature	Importance share	Type
<code>BEL_idiosyncratic_lag1</code>	0.213	abs.coef (normalized)
<code>BEL_idiosyncratic_lag2</code>	0.205	abs.coef (normalized)
<code>PM_nuclear_returns_lag3</code>	0.163	abs.coef (normalized)
<code>PM_nuclear_returns_lag1</code>	0.163	abs.coef (normalized)
<code>PM_nuclear_returns_lag2</code>	0.119	abs.coef (normalized)
<code>BEL_idiosyncratic_lag3</code>	0.099	abs.coef (normalized)
<code>time_to_expiry</code>	0.039	abs.coef (normalized)
<i>Sum</i>	1.001	

Table 6 is important for interpretation. The first two autoregressive lags (`BEL_idiosyncratic_lag1`, share 0.213, and `lag2`, share 0.205) are the two largest individual predictors, with the three own-return lags together accounting for approximately 52% of the normalized total. The three prediction market lags contribute the remaining 44% (lags 1 and 3 equal at 0.163, lag 2 at 0.119), making the PM signal a genuine secondary source of predictive weight rather than a negligible add-on. `time_to_expiry` contributes approximately 3.9%, consistent with

its role as a slowly varying deterministic feature. Note that the raw absolute coefficients do not sum to 1.0 by construction for Ridge (unlike impurity-based tree importances), so the normalisation step described in the methodology is necessary to compare shares across features and cases.

This conclusion is reinforced by the one-step counterfactual sensitivity analysis. When `x_lag1` (the first lag of the nuclear PM return) is varied from -30 to $+30$ percentage points, the predicted next-period Belgian idiosyncratic return changes by approximately 0.09 symmetric-return points across that full range — where the symmetric return is measured on the scale $200 \times (p_t - p_{t-1}) / (|p_t| + |p_{t-1}|)$ — confirming that the fitted linear response is statistically present but economically modest in magnitude.

The model is validated using time-respecting out-of-sample procedures rather than random cross-validation. The final choice of **Ridge** therefore reflects not only strong relative in-sample performance, but also the best-generalizing behavior under temporally ordered model selection (Bergmeir and Benítez 2012; Hyndman and Athanasopoulos 2021).

The main drawbacks of **Ridge** should be stated clearly. First, its linear structure means it cannot capture nonlinear threshold effects or state-dependent responses that may be present when event probabilities approach 0 or 1. Second, the negative out-of-sample R^2 confirms that the model’s predictive contribution is weak: it does not reliably forecast Belgian idiosyncratic returns better than a simple mean forecast on held-out data. Third, Ridge does not perform variable selection: all features receive positive shrinkage rather than exact zeros, making the fitted model less parsimonious than Lasso. Finally, as a linear model it is susceptible to misspecification if the true relationship involves interactions or regime shifts. These limitations are acceptable in the present case study because the objective is scenario generation under a validated linear predictive structure rather than a strongly predictive forecasting system; the Ridge model provides a disciplined, overfitting-resistant baseline for constructing the downstream branching simulation.

These trade-offs accepted, the **Ridge** specification serves as the final forecasting model used to translate shocks in the prediction market series into projected paths for Belgian idiosyncratic electricity returns.

5.2.7 Scenario Simulation and Results

With the final **Ridge** specification in place, we next translate the fitted predictive relationship into scenario-based outputs. The purpose of this stage is twofold. First, we examine how shocks to the prediction market return series map into projected movements in `BEL_idiosyncratic`. Second, we use those projected responses to construct a branching scenario tree and a simple hedge-sizing recommendation. In both steps, the underlying input is the joint time-series structure learned from lagged values of the Belgian idiosyncratic return series and the nuclear PM return series. Accordingly, the outputs should be interpreted as model-implied predictive scenarios rather than as structural causal effects.

We begin with a model-based impulse-response style exercise. Because the final model is a regularized linear Ridge regression, the predictive sensitivity function is smooth and approximately symmetric across shock sizes. The resulting predictive shock-response profiles are shown in Figure 10. The responses are fast-decaying: the largest effects occur at step 1, where positive shocks to the prediction market return generate positive responses

in `BEL_idiosyncratic` and negative shocks generate negative responses proportionally. By step 3, nearly all paths have converged back toward zero, reflecting the modest persistence of the Belgian residual series and the limited lag memory of the fitted model. This clean, quickly reverting response pattern is characteristic of a well-regularized linear predictor in a low signal-to-noise environment, and it stands in contrast to what a more aggressively tuned nonlinear model might produce.

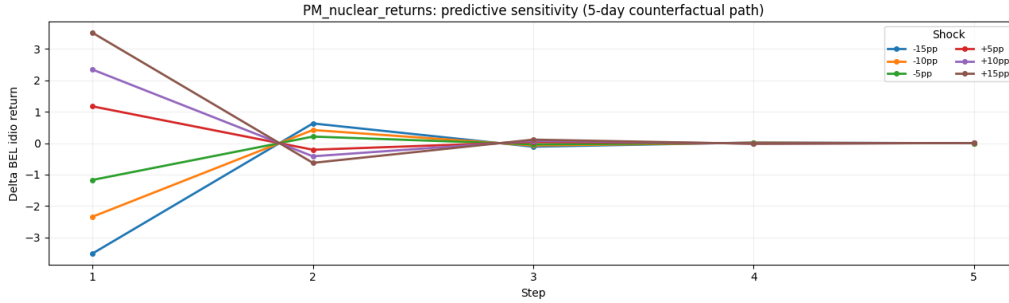


Figure 10: Predictive sensitivity profiles for the final **Ridge** model: 5-day counterfactual paths for `BEL_idiosyncratic` under shocks of ± 5 , ± 10 , and ± 15 percentage points to `PM_nuclear_returns`. Responses are approximately symmetric and decay to near-zero by step 3. Source: author’s calculations.

Before constructing the forward-looking scenario tree, Figure 11 provides retrospective context by plotting the 20-day rolling CVaR_{80} of `BEL_idiosyncratic` alongside the nuclear PM return series over the 2025 sample. The rolling CVaR_{80} averaged approximately -33.9 symmetric-return units ($\times 200$ scale, equivalent to roughly -17% proportional return) over the sample period, with a standard deviation of 17.1 units; the most severe 20-day tail episode reached -89.6 units in the autumn, corresponding to the cluster of large price spikes visible in the raw Belgian electricity series. Periods of elevated CVaR severity (mid-May, late summer, and October–November) broadly coincide with episodes of higher nuclear prediction market return volatility, suggesting that geopolitical risk episodes and electricity-price tail risk cluster in time even if the lead-lag relationship is weak at the daily frequency used throughout the analysis.

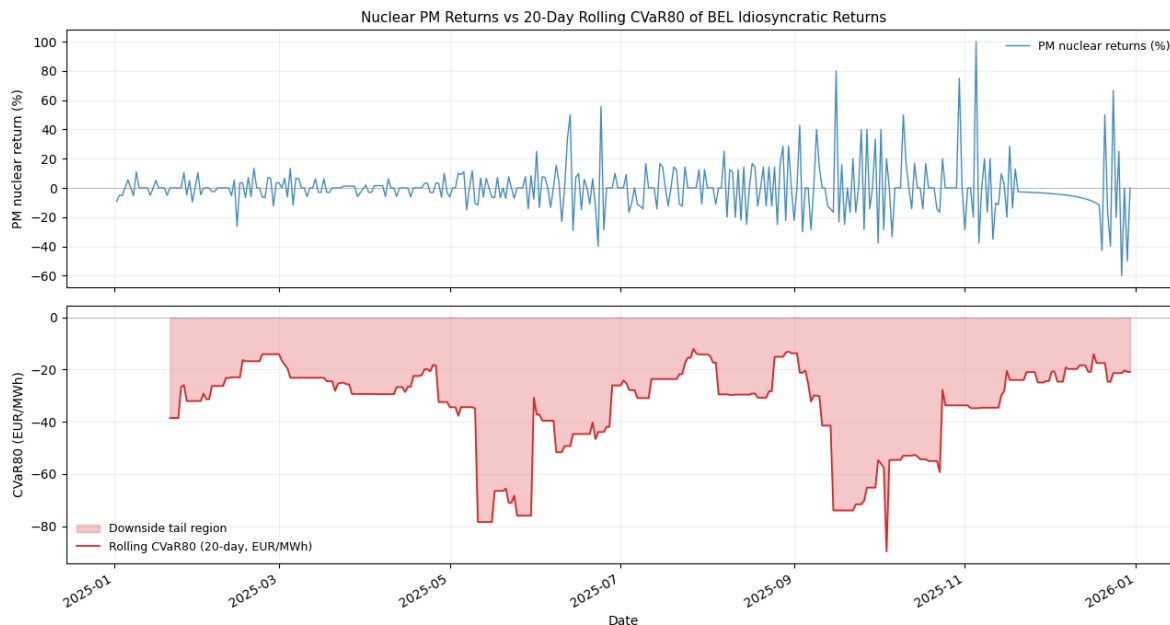


Figure 11: Nuclear prediction market returns (upper panel) and 20-day rolling CVaR_{80} of `BEL_idiosyncratic` (lower panel, symmetric-return units $\times 200$) over 2025. More negative CVaR_{80} values indicate heavier downside tails in that 20-day window. The shaded area marks the downside tail region. Source: author's calculations.

The same logic carries over into the branching simulation. Starting from the selected forecast origin (2025-12-25), the notebook constructs a two-stage empirical branching tree based on the nuclear PM return process and then feeds those paths through the fitted Ridge model to obtain corresponding cumulative abnormal return (CAR) projections for `BEL_idiosyncratic`. The first stage splits the market into an `up` and `down` move with empirically calibrated conditional probabilities (Laplace-smoothed from historical data): the stage-1 probability of an up move is approximately 0.305 and of a down move is 0.695. Each of those branches is then split again at the second stage, yielding four terminal paths with the following calibrated probabilities: up-up (0.077), up-down (0.228), down-up (0.230), and down-down (0.465). The resulting tree is shown in Figure 12.

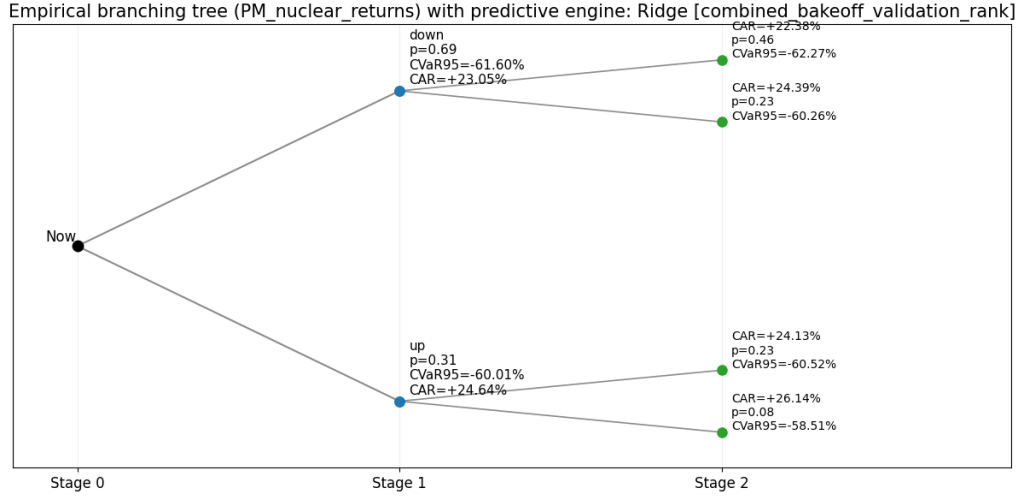


Figure 12: Empirical branching tree for PM_nuclear_returns with model-implied CAR projections for BEL_idiosyncratic. Branch probabilities are calibrated empirically from historical up/down transitions with Laplace smoothing. Source: author's calculations.

Several features of the scenario tree are noteworthy. All four terminal nodes imply positive cumulative abnormal returns for the Belgian idiosyncratic series (ranging from approximately 0.22 to 0.26), reflecting the positive autoregressive structure of the BEL series at the selected forecast origin. The down-resolution state carries a probability of approximately 0.693, while the up state carries 0.307. The dual-tail CVaR values at the terminal nodes show left-tail CVaR₉₅ (adverse for long/producer positions) of approximately -0.60 and right-tail CVaR₉₅ (adverse for short/consumer positions) of approximately $+0.90$, indicating that the consumer-side (short-price) exposure faces a materially larger adverse tail than the producer side at this forecast origin.

Rolling CVaR summary and hedge sizing. Table 7 reports the descriptive statistics for the 20-day rolling CVaR₈₀ series across the full 2025 sample, providing a historical reference for the severity of downside tail risk in the Belgian idiosyncratic return series.

Table 7: Rolling CVaR_{80} summary statistics for `BEL_idiosyncratic` (20-day window, symmetric-return units scaled by 200; divide by 200 to convert to proportional returns)

Statistic	Value
Count	340
Mean	−33.88
Std dev	17.12
Minimum	−89.62
5th percentile	−73.91
25th percentile	−39.53
Median	−29.30
75th percentile	−23.06
95th percentile	−14.19
Maximum	−12.02

The hedge-sizing output is built directly on these terminal scenarios. Given a stated exposure mode, portfolio value, risk budget, execution haircut, and hedge-efficiency parameter, the notebook evaluates whether the scenario-weighted downside implied by the branching simulation exceeds the investor’s tolerance threshold. In this case study, the analysis is run for a `short_price` exposure with a portfolio value of €1,000,000, a risk-budget fraction of 1.5%, an execution haircut of 0.7, and a hedge-efficiency parameter of 0.6. Under the `short_price` convention, the participant is harmed by rising electricity prices, so the loss at each scenario node is $L_s = \max(r_s, 0)$ (from the loss formula in Section ??). All four terminal branches carry positive CARs (0.22–0.26), meaning electricity prices are expected to rise in every scenario: each branch therefore inflicts a positive loss on a short-price consumer. The probability-weighted loss accordingly exceeds the 1.5% risk budget, triggering a positive hedge notional. Table 8 reports the full hedge-sizing recommendation on both a point-estimate (CAR) basis and a CVaR_{95} bootstrap basis, and Table 9 provides the terminal scenario detail including the dual-tail CVaR values.

Table 8: Scenario-based hedge summary: point-estimate vs. CVaR_{95} (bootstrap)

Metric	Point-estimate (CAR)	CVaR_{95} (bootstrap)
Portfolio value	€1,000,000	
Exposure mode	<code>short_price</code>	
Risk-budget fraction	0.015	
Execution haircut	0.70	
Hedge efficiency	0.60	
Expected weighted loss	0.2353	0.8981
Target coverage	0.9363	0.9833
Hedge notional (% exposure)	80.0%	80.0%
Hedge notional (EUR)	€800,000	€800,000
Recommended structure	Severe downside: collar + partial de-risking; larger gas/power downside sleeve or cross-commodity hedge	

Table 9: Terminal scenario detail (Ridge model, forecast origin 2025-12-25)

Stage 1	Stage 2	Prob	CAR	CVaR ₉₅ left-tail	CVaR ₉₅ right-tail
up	up	0.077	+0.261	−0.585	+0.924
up	down	0.228	+0.241	−0.605	+0.904
down	up	0.230	+0.244	−0.603	+0.907
down	down	0.465	+0.224	−0.623	+0.887

The recommendation on both a point-estimate and a CVaR₉₅ basis is a hedge notional of €800,000 (80% of the portfolio value), with a recommended structure of a collar combined with partial de-risking and a larger gas/power downside sleeve or cross-commodity hedge. This is a materially different and more conservative outcome than a no-hedge recommendation, and it reflects the severity of the right-tail risk facing a short-price consumer exposure at this forecast origin: CVaR₉₅ right-tail values across the four terminal branches range from approximately 0.887 to 0.924, far exceeding the 1.5% risk budget on both the point-estimate and bootstrap bases. The convergence of both notionals at the 80% cap reflects that the hedge capacity is exhausted under either input.

More broadly, the results of this subsection illustrate the value of combining a stable, validated linear forecasting model with scenario-based risk analysis. Even with a modest out-of-sample R^2 , the Ridge model produces a scenario distribution that is coherent with the historical tail risk profile documented by the rolling CVaR₈₀ diagnostic, and the dual-tail CVaR output ensures that the hedge recommendation is meaningful for both producer and consumer exposures. The branching tree and hedge recommendation should therefore be read as disciplined predictive tools grounded in validated time-series forecasting, not as stylized deterministic narratives.

5.3 Argentine Equities

5.3.1 Case Context and Model Fitting

The Argentine case study examines whether changes in electoral expectations surrounding the 2025 legislative election help explain short-run movements in Argentine equities. Argentina combines unusually high political uncertainty with a stock market that is highly sensitive to changes in macroeconomic policy, capital controls, and investor confidence. The 2025 midterm election was especially consequential because it was widely viewed as a referendum on President Javier Milei’s reform agenda. Reuters described the vote as a pivotal test of whether Milei could continue his economic overhaul, with investors closely watching whether his party, La Libertad Avanza (LLA), could strengthen its legislative position (Misculin and Miller 2025b; Misculin and Miller 2025a).

This political setting also had direct financial relevance. By mid-2025, foreign and domestic investors had shown renewed interest in Argentina following Milei’s program of fiscal adjustment, deregulation, and partial relaxation of foreign-exchange restrictions, which had revived deal activity and market sentiment (Raszewski and P. Henderson 2025). At the same time, the political environment remained polarized, making the electoral outlook relevant not only for politics but also for asset pricing (Misculin and Miller 2025b).

To construct the target variable, MERV returns are residualized against a PCA factor block built from ex-Argentina South American proxy ETFs (ILF, EWZ, ECH, EPU, GXG), so that the residual `MERV_idiosyncratic` captures the country-specific component of Argentine equity returns. Note that unlike the Belgian electricity case, MERV returns are expressed as plain percentage changes rather than the symmetric return transform, because the Mervel index is always positive and far from zero. The LLA prediction market contract resolves on 2025-11-19, when the election outcome is known with certainty, so the time-to-expiry variable counts down from the start of the sample to that date.

Before proceeding to model fitting, we verify stationarity. For `LLA_returns` the ADF test fails to reject the unit-root null (stat = 0.078, $p = 0.965$) while the KPSS test also fails to reject stationarity (stat = 0.202, $p = 0.100$) — a conflict consistent with a slowly drifting prediction market series approaching a known binary resolution. For `MERV_idiosyncratic`, both tests agree on stationarity (ADF: stat = -4.658 , $p < 0.001$; KPSS: stat = 0.308, $p = 0.100$). Given the ambiguous stationarity of `LLA_returns`, a Johansen trace test is applied to assess cointegration. The trace test rejects no cointegration (stat = 35.35 > CV = 15.49) and also rejects rank $r \leq 1$ (stat = 10.32 > CV = 3.84), yielding an inferred rank of 2. Because full rank implies the system is already stationary in levels, the VECM is excluded on statistical grounds and the notebook selects VAR(1) by BIC (BIC₁ = -12.628 vs. BIC₂ = -12.526 ; a marginal preference for lag 1) as the traditional benchmark instead.

The expanding-window backtest across 168 observations reveals that the ML and Traditional families perform nearly identically at the aggregate level: ML best RMSE is 0.035 against Traditional best RMSE of 0.035 as well, with all directional accuracies clustered around 0.46–0.51. The best individual model overall is `ElasticNet` (RMSE = 0.035, MAE = 0.023, OOS $R^2 = -0.009$), followed closely by `Ridge`. The negative OOS R^2 values across almost all models are notable: at the expanding-window horizon no specification reliably outperforms a simple mean forecast over the full sample.

Table 10: Expanding-window backtest ranking for the Argentine case (168 observations, expanding window starting at train size 50)

Model	Family	RMSE	MAE	OOS R^2	Dir. acc.
ElasticNet	ML	0.0347	0.0229	-0.009	0.483
Ridge	ML	0.0349	0.0230	-0.020	0.458
GradientBoosting	ML	0.0354	0.0237	-0.054	0.441
ExtraTrees	ML	0.0355	0.0235	-0.054	0.500
VAR	Traditional	0.0355	0.0234	-0.056	0.458
RandomForest	ML	0.0363	0.0240	-0.106	0.475
SVR	ML	0.0418	0.0295	-0.468	0.508

A key empirical finding is that model performance is highly sensitive to training-window length. When the window-grid search is run across all model-window combinations, the best specifications cluster around 148–163 trading days, with `ElasticNet` at 148 days achieving RMSE = 0.019 and OOS $R^2 = 0.300$ — far superior to any result from the full-sample expanding window. Figure 13 illustrates this pattern. The strong window-sensitivity implies

that the relationship between LLA expectations and Argentine equities is not stable over the full year but becomes more informative in a narrower window as the election approaches.

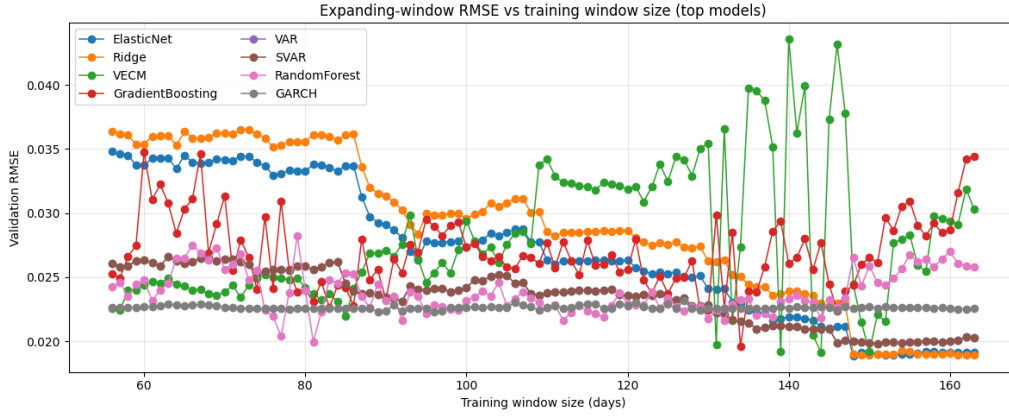


Figure 13: Expanding-window RMSE by training-window size for the strongest candidate models. The best-performing specifications cluster in windows of approximately 148–163 days, suggesting that predictive structure strengthens as the election approaches. Source: author’s calculations.

Table 11: Best model-window combinations for the Argentine case

Model	Family	Window (days)	Lags	RMSE	OOS R^2
ElasticNet	ML	148	3	0.01885	0.300
Ridge	ML	150	3	0.01891	0.295
ElasticNet	ML	153	3	0.01893	0.294
VECM	Traditional	144	2	0.01915	0.277
GradientBoosting	ML	134	3	0.01961	0.242

The combined bakeoff-plus-validation ranking selects **ElasticNet** as the final simulation model: it ranks first in the bakeoff and first in the validation task, giving a combined rank score of 2.0, ahead of Ridge (4.0) and VECM (6.0). The simulation training window is the best-performing 148-day window, and the model reports an in-sample R^2 of 0.062 and a simulation-holdout OOS R^2 of -0.020 on the five observations beyond the anchor date. The near-zero and negative OOS metrics confirm that while the ElasticNet identifies some structure within its 148-day window, its out-of-sample predictive contribution at the forecast anchor is very limited.

The feature-importance output reinforces this conclusion. Computed using absolute coefficient values on the held-out 20% of the training window, the largest coefficient belongs to `y_lag2` (-0.0037), followed by `tte_current` ($+0.0027$) and `x_lag2` (-0.0016). Crucially, the ElasticNet regularization has zeroed out the coefficients on `x_lag1`, `tte_lag1`, `tte_lag2`, `x_lag3`, and `y_lag3` entirely. This means the LLA prediction market signal is not selected by the fitted model as a reliable predictor at this forecast origin: the regulariser effectively

decides that the incremental information in lagged LLA returns does not justify the associated variance. The counterfactual sensitivity sweep over `x_lag1` in $[-30\%, +30\%]$ confirms this, with the predicted next-period `MERV_idiosyncratic` changing by only 0.037 across the full range.

5.3.2 Scenario Outputs and Takeaways

The selected `ElasticNet` model with its 148-day window is used in the branching simulation. The simulation anchors at 2025-11-12 (T−5), five trading days before the election resolution on 2025-11-19, with the MERV index at 2,984,608. At this anchor, the LLA level is 0.72, and the stage-1 binomial calibration assigns a down-node probability of 0.71 and an up-node probability of 0.73, reflecting the high probability of LLA winning already priced into the market near resolution.

The terminal scenario summary shows that all four branches produce nearly identical positive cumulative abnormal returns of approximately +0.060 to +0.062, with zero modeled drawdown in every case. The two terminal resolution states carry probabilities of 0.28 (resolve to 0, LLA does not win outright) and 0.72 (resolve to 1, LLA wins), yet the expected terminal abnormal returns are nearly identical at approximately 0.060961 and 0.060987 respectively.

Table 12: Terminal scenario summary for the Argentine case (`ElasticNet`, 148-day window, anchor 2025-11-12)

Resolution	Prob	CAR	CVaR ₉₅ long-tail	CVaR ₉₅ short-tail	Max drawdown
Resolves to 0	0.28	+0.061	+0.001	+0.006	0.000
Resolves to 1	0.72	+0.061	+0.000	+0.005	0.000

The hedge-sizing output follows directly from this near-flat scenario distribution. Because all terminal branches produce positive CARs and the weighted downside loss is zero, both the point-estimate and CVaR₉₅ bootstrap hedge notionals are 0. The recommended structure is no additional hedge.

The rolling CVaR₈₀ diagnostic provides retrospective context. Over 149 rolling windows, the 20-day rolling CVaR₈₀ of `MERV_idiosyncratic` averages −0.033 with a standard deviation of 0.013, ranging from a minimum of −0.059 to a maximum of −0.012. These are dimensionless return proportions consistent with the plain `pct_change` scale used for Argentine equities. The severity is broadly comparable in relative terms to the Belgian case but without the systematic co-movement with PM return volatility that characterized the nuclear case.

Relative to the Belgian electricity analysis, the Argentine case therefore produces a markedly weaker hedging signal. Three features account for this. First, the regularization entirely suppresses the LLA PM lags in the fitted `ElasticNet`, meaning the model assigns no predictive weight to the political signal at this window. Second, all scenario branches converge to nearly identical positive returns, reflecting the autoregressive momentum in the MERV series near election day rather than any differential political sensitivity. Third, the

election resolution was highly predictable five days out — LLA was already at 0.72 — leaving little scenario spread to exploit.

The case is nonetheless informative as a negative result with a structural interpretation. When event expectations are largely resolved, as they were in Argentina five days before a widely anticipated election outcome, the PM provides no incremental hedging signal beyond what the equity market has already priced. The earlier window-sensitivity finding points to a more tractable use case: at greater distance from resolution, shorter training windows that capture the period of rising political salience do identify a meaningful relationship. This suggests that the framework’s value in event-driven equity settings may be concentrated in an intermediate salience window — close enough to election day for the political factor to matter, but far enough out for genuine uncertainty to remain (Bauer and T. M. Mertens 2023; Engelberg, Reed, and Ringgenberg 2011).

5.4 Artificial General Intelligence Expectations

5.4.1 Background and Data

This final case study considers whether market expectations about artificial general intelligence (AGI) help explain returns on an AI-themed equity investment. Although the term remains contested, AGI is usually understood as AI with broad, human-level capability across many tasks rather than strong performance in one narrow domain (Hendrycks et al. 2025). This topic is financially relevant because expectations about the future usefulness of AI have already shaped the performance of technology equities. Recent work in finance documents both valuation-driven enthusiasm around AI-related assets and herding behavior in generative-AI stocks following major technological milestones (Bergsma 2024; Wang 2024).

The financial asset is the ARTY ETF (iShares Future AI & Tech ETF), whose portfolio is concentrated in the hardware, semiconductor, and compute layer of the AI ecosystem: recent holdings include CoreWeave, Marvell, AMD, TSMC, Broadcom, Micron, NVIDIA, and Oracle (BlackRock 2026; Stock Analysis 2026). Data run from April 2024 through end-2025, giving 440 observations in total.

A key difference from the previous case studies is the absence of a cash-settled PM for the exact outcome of interest. Instead, the AGI signal is constructed from the AGI Timelines Dashboard maintained by Good Heart Labs, which aggregates four forecasting series — three Metaculus timeline questions and one Manifold series — into a common AGI-expectation measure transformed by log differences (Good Heart Labs 2026). This introduces several important limitations: the underlying questions are similar but not identical, they evolve slowly (most daily changes are exactly zero, as visible in the distribution of the AGI series in Figure 14), and they come from platforms with weaker monetary incentives than regulated cash-settled markets (Servan-Schreiber et al. 2004; Metaculus 2025).

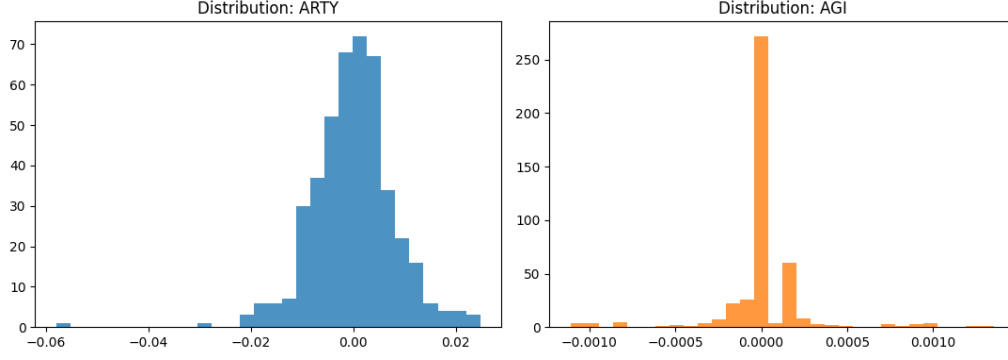


Figure 14: Empirical distributions of **ARTY** idiosyncratic returns (left) and the AGI log-difference signal (right). The AGI series is extremely sparse: the overwhelming majority of daily observations are exactly zero, with only occasional small non-zero shifts. This structural sparsity is the primary constraint on any predictive relationship between the two series. Source: author’s calculations.

The factor model residualises **ARTY** returns against a PCA block built from the Nasdaq-100 ($\sim$ NDX), global equities (ACWI), and long rates ($\sim$ TNX). The SVD is fitted on 420 training observations and produces two retained components explaining 65.2% (PC1) and 32.8% (PC2) of factor variance, with a third component explaining only 2.0% and thus excluded. The Huber robust regression of **ARTY** returns on these two components leaves a residual series **ARTY_idiosyncratic** with a mean of -0.000044 , standard deviation of 0.0081 , skewness of -0.76 , and excess kurtosis of 6.14 — confirming the expected heavy-tailed structure.

5.4.2 Stationarity, Cross-Correlations, and Structural Diagnostics

Both series pass the stationarity battery decisively. The ADF test (-20.78 , $p < 0.001$ for **ARTY**; -8.73 , $p < 10^{-13}$ for **AGI**), Phillips–Perron (-20.90 , $p = 0.0$; -37.25 , $p = 0.0$), and Zivot–Andrews tests (-20.94 , $p < 0.001$; -10.40 , $p < 0.001$) all reject the unit-root null for both series. The KPSS statistics of 0.118 and 0.138 respectively fail to reject stationarity (both $p = 0.10$, at the boundary of the look-up table). The Zivot–Andrews break is identified at index 0 for **ARTY** and index 10 for **AGI**, both at the very start of the sample, consistent with the early high-volatility episode in the **ARTY** ETF’s trading history rather than a mid-sample structural shift. As shown in Figure 15, the three estimated breakpoints from the change-point detection algorithm cluster in May–July 2024 and correspond to that initial volatile episode; a CUSUM test on the OLS residuals of **ARTY** on the **AGI** signal confirms no drift in the fitted relationship itself ($p = 0.662$).

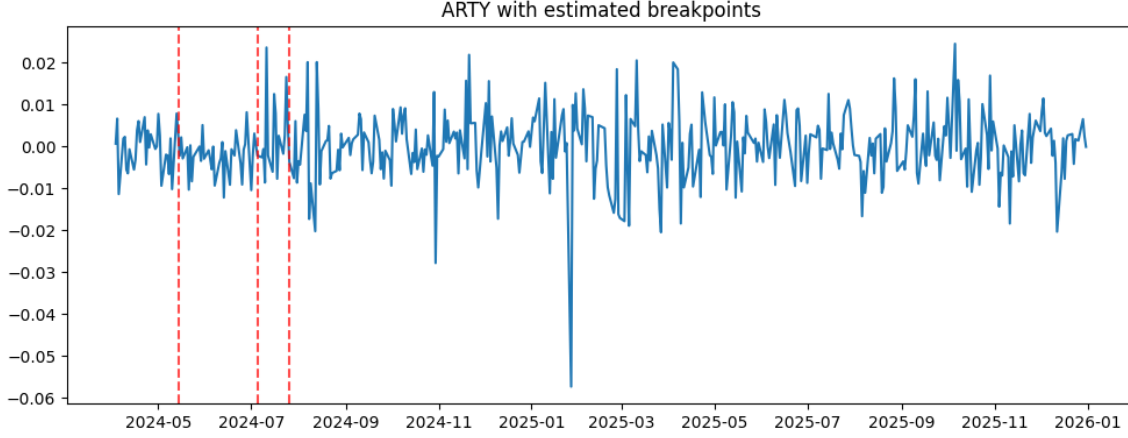


Figure 15: ARTY idiosyncratic return series with estimated breakpoints (red dashed lines). The three estimated breaks cluster in May–July 2024, coinciding with an early high-volatility episode at the start of the available sample. Source: author’s calculations.

The cross-correlation analysis finds no meaningful lagged relationship between the AGI signal and ARTY returns. The largest raw CCF is -0.054 at lag 6; the largest prewhitened CCF is 0.061 at lag 16; all remaining values are below 0.05 in absolute magnitude. An ARDL model comparison by BIC selects $ARDL(p=1, q=0)$ — meaning no AGI lags are retained even in a flexible single-equation distributed-lag framework. The distributed-lag sum is $\hat{\beta} = 0.250$ ($SE = 0.940$, $p = 0.790$), statistically indistinguishable from zero. The Johansen trace test on the bivariate system yields rank 2 ($stat = 189.2 > CV = 15.49$ for $r \leq 0$; $stat = 74.2 > CV = 3.84$ for $r \leq 1$), implying the system is already stationary in levels and excluding a VECM; $VAR(3)$ is selected by AIC ($AIC = -26.268$). The Engle ARCH LM test at lag 10 finds no evidence of volatility clustering in the ARTY residual ($LM = 4.79$, $p = 0.905$; $F = 0.472$, $p = 0.908$), ruling out misspecified conditional variance as an explanation for the poor fit.

The Local Projections IRF, shown in Figure 16, corroborates this picture. The estimated response of ARTY to a unit AGI shock is indistinguishable from zero at all horizons from $h = 0$ to $h = 10$. The only nominally positive estimate is at $h = 9$ ($\hat{\theta}_9 = 2.455$, $SE = 1.034$, 95% CI $[0.429, 4.482]$), but given 11 simultaneous tests and the reversion at $h = 10$, this should not be interpreted as a robust finding.

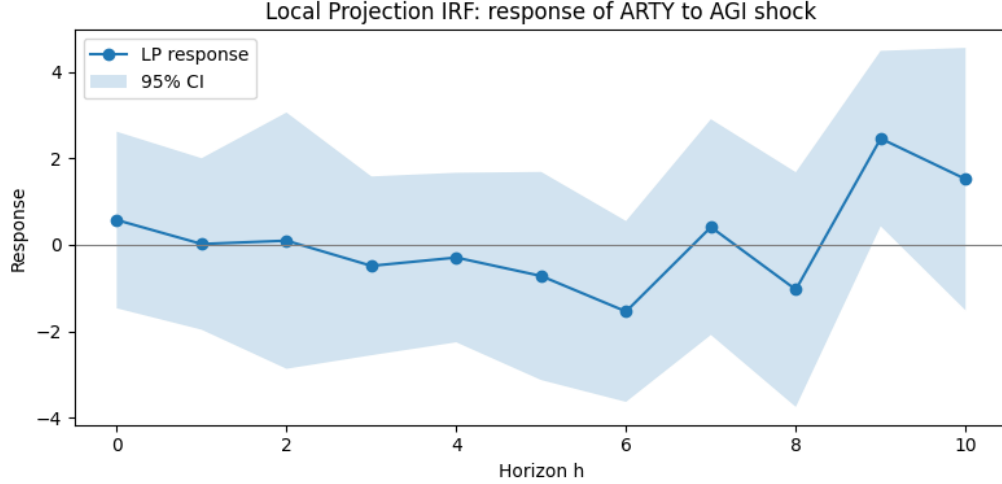


Figure 16: Local Projections impulse response function: estimated response of **ARTY** idiosyncratic returns to a unit AGI log-difference shock, with HAC-robust 95% confidence intervals. The response is indistinguishable from zero at all horizons. Source: author’s calculations.

5.4.3 Model Selection and Scenario Output

The expanding-window backtest over 293 forecasts selects **ElasticNet** as the best model (RMSE = 0.00873, MAE = 0.00638, OOS R^2 = -0.006), trained on 439 observations with three lags. Table 13 reports the full backtest ranking. The performance differences across all model families are negligible: The Traditional benchmark (VAR) achieves RMSE = 0.00882 and GARCH achieves 0.00896, barely worse. A rolling 50-step R^2 evaluation confirms this: mean = -0.024 , median = -0.014 , with the 10th–90th percentile range of $[-0.061, -0.004]$ entirely negative. No specification generates reliable positive predictive signal at any horizon.

Table 13: Expanding-window backtest results for the AGI case (293 forecasts)

Model	Family	RMSE	OOS R^2	Dir. acc.
ElasticNet	ML	0.008731	-0.006	0.413
Ridge	ML	0.008731	-0.006	0.410
GradientBoosting	ML	0.008777	-0.017	0.515
VAR	Traditional	0.008819	-0.027	0.505
GARCH	GARCH	0.008963	-0.061	0.451
ExtraTrees	ML	0.009133	-0.101	0.526
RandomForest	ML	0.009249	-0.129	0.505

The selected **ElasticNet** model has an in-sample R^2 of 0.0006 (effectively zero) and an OOS R^2 of -0.006 . The feature-importance output makes the situation concrete: using absolute coefficients evaluated on the holdout 20% of the training window, the only non-negligible coefficient belongs to `y_lag3` (-0.000516), while every other feature — including `x_lag1`, `x_lag2`, `x_lag3`, `y_lag1`, and `y_lag2` — has a coefficient of exactly zero. The one-step counterfactual sensitivity sweep over `x_lag1` in $[-30\%, +30\%]$ returns a delta of 0.0

across every tested value, confirming that the AGI signal has been entirely eliminated by regularization and produces literally no predictive variation in the fitted model.

The scenario tree, shown in Figure 17, uses three stage-1 AGI-timeline regimes (**advanced** at probability 0.25, **central** at 0.50, **severe_delay** at 0.25) and three stage-2 sub-branches each, giving nine terminal paths over a 10-day horizon. The stage-1 shocks correspond to the 10th, 50th, and 90th percentiles of observed 10-step AGI shifts (-1.08 , 0.0 , and $+1.25$ years respectively). Despite these materially different AGI-timeline trajectories, every terminal branch produces an identical CAR of 0.000128 , with a zero counterfactual delta at every node (Table 14). The branching is purely mechanical: the regularized model ignores the AGI shocks entirely and outputs the same near-zero value regardless of branch.

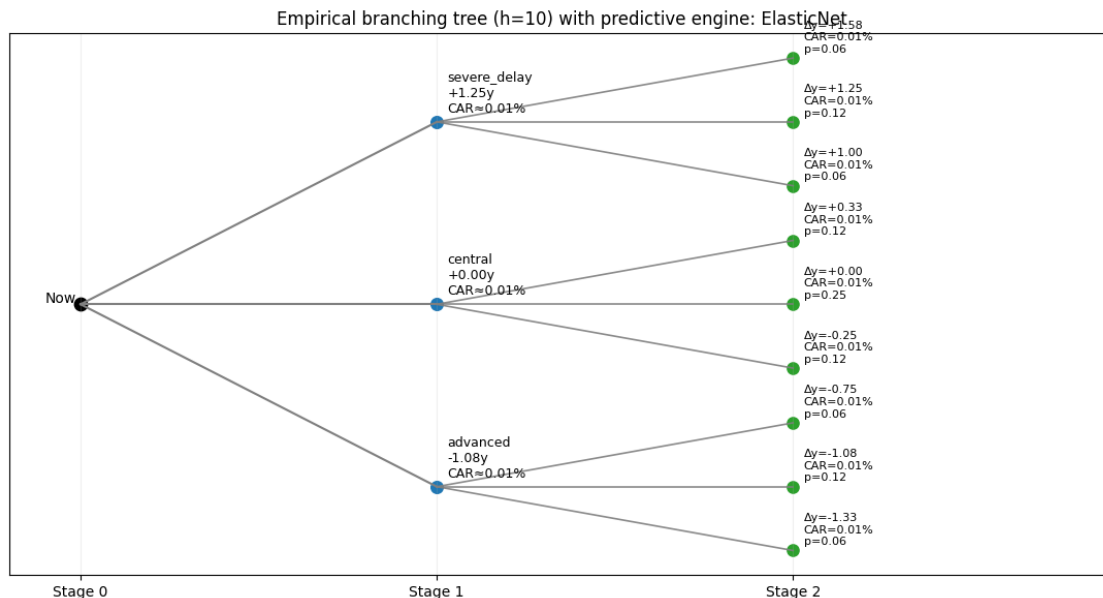


Figure 17: Branching simulation for the AGI case using the selected ElasticNet model with a 10-day horizon. Despite materially different AGI-timeline regimes across the nine terminal branches, all CARs are identical at approximately $+0.01\%$, reflecting complete regularization-driven elimination of the AGI signal. Source: author’s calculations.

Table 14: Terminal scenario statistics by AGI stage (ElasticNet, 10-day horizon)

Stage	Prob	Min CAR	Mean CAR	Max CAR
Advanced AGI	0.25	0.000128	0.000128	0.000128
Central AGI	0.50	0.000128	0.000128	0.000128
Delayed AGI	0.25	0.000128	0.000128	0.000128

The hedge-sizing output produces a striking two-way result. On the point-estimate (CAR) basis, all scenario branches are positive, the probability-weighted downside is zero, and the hedge notional is 0. However, the bootstrap CVaR₉₅ calculation — which draws from OOS 80/20 residuals of the lagged feature matrix via block bootstrap — produces a weighted

CVaR loss of 0.053, far exceeding the 1.5% risk budget. This drives a CVaR-informed hedge notional of 800,000 (80% of the portfolio value, capped). This divergence between the point-estimate and CVaR-informed notionals is itself an important result: it reflects the fact that while the model’s point predictions are near-zero and positive on average, the tail distribution of the bootstrap residuals is heavy enough to warrant significant protection if the tail materialises. In other words, the hedge demand is driven entirely by the uncertainty in the innovation distribution, not by any directional prediction market signal. This is the opposite of the Belgian electricity case, where the point-estimate already generated a material hedge demand.

5.4.4 Interpretation

The AGI case is best understood as a boundary case for the methodology, and the weight of diagnostic evidence consistently points in one direction. The CCF analysis finds no cross-correlation at any lag. The ARDL model retains no AGI lags by BIC. The Local Projections IRF is flat with overlapping zero confidence intervals at almost every horizon. The ARCH test rules out volatility misspecification. The ElasticNet zeros out every predictor feature. The counterfactual sensitivity is literally zero. Together these results provide a far stronger evidential basis for the null-signal conclusion than the backtest R^2 alone.

The non-result reflects a combination of structural features specific to this case: the AGI signal is constructed from slow-moving, weakly-incentivized play-money forecasting platforms; the series is extremely sparse (the large majority of daily observations are zero); there is no cash settlement or binary resolution forcing prices to carry information; and the 440-day sample, while longer than the original case description suggested, still begins with a heterogeneous high-volatility episode that complicates training. The CVaR-informed 800,000 hedge notional warrants careful interpretation. It is important to note that this notional would fire identically even if the AGI signal were removed from the model entirely: every diagnostic in this case study confirms zero PM contribution (zero CCF, zero LP IRF, zero feature coefficients, zero counterfactual sensitivity). The CVaR hedge is therefore driven solely by the heavy-tailed innovation distribution of the ARTY idiosyncratic residual series, not by any PM-informed directional signal. This distinguishes the AGI CVaR result fundamentally from the Belgian electricity case, where the CAR-based notional was driven by the PM-predicted scenario distribution. A practitioner relying on the CVaR augmentation alone, without inspecting the feature importances and counterfactual sensitivity, could incorrectly infer that the framework has detected an AGI-related tail risk when in fact it has detected a property of the ARTY return distribution that has nothing to do with the prediction market. The AGI case therefore illustrates both a failure mode of the framework (when the PM is structurally unsuited to the task) and a limitation of the CVaR augmentation: it can flag residual tail risk even when the PM contributes nothing, and its output should always be interpreted alongside the feature importance and counterfactual sensitivity diagnostics rather than in isolation (Servan-Schreiber et al. 2004).

6 Discussion of Results

6.1 Cross-Case Performance Comparison

The three case studies were designed to evaluate the proposed framework under materially different conditions: a conflict-linked commodity market at daily frequency, a political election market with time-varying salience, and a technology-expectations signal from non-cash forecasting platforms. Taken together, their results do not support a single uniform verdict on the framework’s usefulness. Instead, they reveal a structured gradient: the Belgian electricity case delivers the clearest and most complete output, the Argentine case is informative about the timing of event relevance despite weak average signal, and the AGI case constitutes a clean null result that delineates the boundary conditions under which the approach fails. The following subsections assess these differences along three shared axes: predictive fit and directional validity; scenario interpretability and decision relevance; and the behavior of model families across cases.

6.1.1 Predictive Fit and Directional Validity

Table 15 summarises the key quantitative outputs across the three case studies. Several features of this comparison deserve attention.

Table 15: Cross-case performance summary. OOS R^2 is the expanding-window backtest statistic for all three cases (Belgium: 118 forecasts; Argentina: 118 forecasts; AGI: 293 forecasts), computed as $1 - \text{SSE}/\text{SST}$ over the full holdout sequence. RMSE units differ: Belgium uses the symmetric-return scale (multiplied by 200, so RMSE 22.88 corresponds to approximately 11.4 symmetric-return units per step, or roughly an 11% return error in the underlying price series), while Argentina and AGI use natural proportional decimal returns. Directional accuracy is the fraction of one-step-ahead forecasts with the correct sign. Hedge notional is the recommended coverage under the stated exposure convention at the single simulation forecast origin.

	Belgium	Argentina	AGI
Selected model	Ridge	ElasticNet	ElasticNet
Training window	Full sample	148 days	Full sample
Sample size (N)	363	168	440
Backtest OOS R^2	−0.079	−0.009	−0.006
Best window OOS R^2	n/a	+0.300	n/a
Best RMSE	22.88	0.019	0.0087
Directional accuracy	0.579	0.483	0.413
Scenario separation	Moderate	Minimal	None
Hedge (CAR basis)	€800,000	€0	€0
Hedge (CVaR basis)	€800,000	€0	€800,000

The OOS R^2 figures are computed on a common expanding-window basis across all three cases. All three values are negative or near zero, which deserves careful interpretation. A

negative OOS R^2 means the selected model performs worse than a naive mean forecast over the full holdout sequence — not that the model is useless in absolute terms, but that whatever predictive signal exists is too weak or too concentrated in short time windows to manifest as consistent one-step-ahead accuracy over the full sample. The Belgian value of -0.079 is negative despite the Ridge model winning the validation task, reflecting that even the best-generalizing specification cannot beat a mean forecast on average. The Argentine value of -0.009 is similarly negative for the full-sample backtest, but the window-grid search reveals that an ElasticNet model trained on the best 148-day window achieves an OOS R^2 of $+0.300$ — a substantial improvement that is invisible in the aggregate backtest statistic. The AGI value of -0.006 is negative and has no analogous window-sensitivity improvement: the rolling 50-step R^2 averages -0.024 with a 10th–90th percentile range of $[-0.061, -0.004]$, entirely negative.

Directional accuracy presents a similarly subdued picture, with Belgium performing best at 0.579 (Ridge, expanding validation task), Argentina next at 0.483 (ElasticNet, full expanding-window backtest), and AGI weakest at 0.413 (ElasticNet, full expanding-window backtest).² Notably, AGI’s directional accuracy is below the 0.50 coin-flip benchmark, confirming that the ElasticNet model’s near-zero predictions actively misalign with the direction of the actual series on average. All figures are consistent with the general finding in the literature that one-step-ahead financial return forecasting is very difficult and that even statistically significant relationships tend to manifest as small, noisy predictive advantages rather than clear directional signals (Pesaran and Timmermann 1992).

One important caveat applies to the RMSE column. Belgian idiosyncratic electricity returns are expressed in symmetric-return units scaled by 200 , while Argentine and AGI returns are in proportional decimal form. Direct numerical comparison is uninformative; what matters is each figure relative to the scale and volatility of the underlying series.

Taken together, the predictive fit results establish a clear ranking: Belgium has the strongest directional accuracy and the most useful window-specific predictive signal (evidenced by the CVaR-informed hedge demand), Argentina is informationally ambiguous but yields a positive in-window OOS R^2 that is invisible in aggregate, and AGI is a uniform null. This ordering motivates the asymmetric treatment of the three cases in the remainder of the discussion.

6.1.2 Scenario Interpretability and Decision Relevance

Predictive fit is a necessary but not sufficient condition for the framework to generate useful hedging signals. A model may achieve positive window-specific R^2 without producing a branching scenario tree with sufficient terminal separation to inform hedge sizing. The three cases illustrate both how scenario separation arises and when it fails.

In the Belgian electricity case, the two-stage empirical branching tree yields four terminal nodes with cumulative abnormal return (CAR) projections ranging from approximately

²Note that the Belgian figure comes from the expanding-window validation task (Task 2), while the Argentine and AGI figures come from the full expanding-window backtest. The two procedures differ in minimum training window size and stride length, so the Belgian directional accuracy is not strictly comparable to the other two on a like-for-like basis. The ordering (Belgium best, AGI worst) is robust to this procedural difference, but the precise gap between Belgium and Argentina should be interpreted cautiously.

+22% to +26%. All four branches point in the same direction (positive), reflecting predominantly autoregressive momentum in the Belgian residual series at the December 2025 forecast origin rather than PM-driven separation between scenarios. The 4 percentage-point spread across branches is therefore better interpreted as quantifying *how strongly* the model-implied upward momentum differs across PM paths than as a case where PM shocks generate qualitatively different outcomes. The hedge-sizing output is nonetheless materially non-zero: under the **short_price** convention, all positive CARs represent losses for a consumer, so the probability-weighted loss exhausts the risk budget and both the CAR-based and CVaR₉₅-based notionals reach the 80% cap at 800,000. The more compelling evidence for PM-signal contribution in the Belgian case comes from the feature importance analysis (44% PM share in the fitted Ridge model) rather than from the single-origin scenario tree, and a different forecast origin — at a period of genuine PM-driven uncertainty rather than year-end momentum — might produce greater between-branch separation.

The Argentine case produces near-flat terminal returns across the two political resolution states: approximately +6.10% in both the resolve-to-0 and resolve-to-1 branches, a difference of less than 0.3 basis points. The probability-weighted downside under the long-price exposure convention is zero, and the hedge recommendation is 0 on both the CAR and CVaR bases. This outcome is structurally different from the Belgian result: in Argentina, the scenario tree is flat not because the model is uncertain but because the fitted ElasticNet model has zeroed out the LLA prediction market lags entirely, leaving only near-zero autoregressive coefficients. The model is essentially predicting a small positive constant return regardless of the PM path, so the branches are indistinguishable. In economic terms, the framework correctly concludes that there is no model-implied downside risk to hedge five trading days before a widely anticipated election outcome — which, at an LLA level of 0.72, was already largely resolved.

The AGI branching tree is completely flat for the same reason at a deeper level. Despite three distinct AGI-timeline scenario paths (*advanced*, *central*, *severe delay*), the selected ElasticNet model assigns the same terminal CAR of approximately +0.013% to all nine terminal branches. The hedge-sizing routine returns 0 on the CAR basis. However, the CVaR₉₅ bootstrap produces a hedge demand of 800,000 — not because the model identifies any AGI-driven downside signal, but because the bootstrap distribution of the ElasticNet’s own residuals is heavy-tailed enough to generate a large tail loss that exceeds the 1.5% risk budget. This is an important distinction: the CVaR hedge demand in the AGI case is driven entirely by innovation-distribution tail risk, not by any directional prediction market signal.

These three scenario outcomes reveal a nuanced taxonomy of framework utility. When the PM signal generates enough variation across branches and the bootstrap residuals are heavy-tailed, both the CAR-based and CVaR-based notionals fire (Belgium). When the PM signal is zeroed out and the residual distribution is relatively thin-tailed, both notionals are zero (Argentina). When the PM signal is zeroed out but the residual distribution is still heavy-tailed — as in the AGI case — the CVaR notional fires even though the CAR notional does not. This last pattern highlights a limitation of the CVaR augmentation: it captures uncertainty in the innovation distribution independently of any PM signal. A CVaR-triggered hedge notional in the absence of PM predictive content (as in the AGI case) reflects the tail properties of the residual series, not an event-driven risk, and should be interpreted alongside the feature importance and counterfactual diagnostics before being

acted upon.

6.1.3 Model Family Behavior Across Cases

A notable feature of Table 15 is that the winning model family is regularized linear in all three cases. Belgium selects Ridge, Argentina selects ElasticNet, and AGI selects ElasticNet. This convergence is itself informative. It suggests that, on the available daily-frequency data, the predictive relationships between prediction market returns and financial asset returns are best captured by linear shrinkage estimators rather than by flexible nonlinear ensemble methods.

In the Belgian electricity case, the model bakeoff shows that SVR ranks first by RMSE and SVR_linear ranks second, with Ridge third — so the top three models are all linear or kernel-based, not tree ensembles. GradientBoosting and RandomForest fall to ranks 10 and 11. The subsequent generalization validation confirms Ridge as the outright winner, with ExtraTrees the best nonlinear model but still behind the regularized linear cluster. This reverses the earlier finding from the ExtraTrees era and carries a substantive interpretation: the relationship between lagged nuclear prediction market returns and Belgian idiosyncratic electricity returns is approximately linear and stable across forecast origins. Whatever threshold or interaction effects tree-based methods might capture in-sample do not persist out-of-sample. The selection of Ridge implies that shrinkage toward a small linear mapping from PM lags and own-return lags to next-period returns is the most defensible specification given the data.

In the Argentine case, ElasticNet ranks first in both the bakeoff and the validation task, with Ridge a close second. GradientBoosting, which was the winner in the previous notebook version, falls to rank 4 in the updated backtest. The convergence toward regularized linear models is consistent with the Belgian finding, but the mechanism differs: in Argentina, the ElasticNet regulariser zeros out almost all feature coefficients, effectively producing a near-constant forecast. This is not a failure of the linear specification; it is the correct output of a well-regularized model applied to a signal that, at daily frequency and over the full 168-observation sample, contains very little usable predictive content. The observation that shorter training windows yield positive OOS R^2 values is consistent with a nonstationary relationship that is not captured by any fixed-parameter model estimated over the full year.

In the AGI case, ElasticNet is again selected, and again regularization zeros out all feature weights. The near-equivalence of all nine candidate models — with RMSE values spanning only 0.00873 to 0.01108 and OOS R^2 values uniformly negative — confirms that no model family can extract usable signal from this data. When the signal-to-noise ratio is sufficiently low, model family becomes irrelevant.

Taken together, the cross-case model behavior is consistent with the no-free-lunch perspective on forecasting (Hyndman and Athanasopoulos 2021): flexible nonlinear models add value only when there is genuine nonlinear structure to exploit, and in its absence regularized linear alternatives perform as well or better. The three cases provide evidence for a gradient between these regimes, with Belgium showing the strongest signal that still falls short of justifying tree-ensemble complexity, and Argentina and AGI confirming that a flat ElasticNet is the appropriate response when daily-frequency predictive structure is weak or absent.

6.2 The Role of Prediction Markets as Predictive Features

A central question for the practical usefulness of the framework is whether prediction markets contribute genuine incremental information about asset returns, over and above what the asset’s own recent history already contains. The feature-importance outputs from the three case studies allow this question to be examined directly.

6.2.1 Prediction Market Lags versus Own-Return Lags

Table 16 reports the feature importances for the final simulation model in each case study, together with the implied share of total importance attributable to the prediction market lags and to the asset’s own lagged returns. All three models use absolute coefficient values evaluated on the held-out 20% of the training window.

Table 16: Normalized feature importance shares for the final simulation model in each case study. All values are absolute coefficient magnitudes from the Ridge or ElasticNet specification, evaluated on the training window and normalized to sum to 1.0 (following the code fix described in Section ??, which clips negative permutation scores and divides by the absolute sum). Dashes indicate features zeroed by regularization. PM share sums x -lag shares; own-ret. share sums y -lag shares; TTR is the time-to-expiry/resolution feature.

	PM lags (x)			Own-return lags (y)			TTR	PM share	Own-ret. share
	lag 1	lag 2	lag 3	lag 1	lag 2	lag 3			
Belgium (Ridge)	0.163	0.119	0.163	0.213	0.205	0.099	0.039	44.4%	51.7%
Argentina (ElasticNet)	—	0.178	—	0.117	0.411	—	0.294	17.8%	52.8%
AGI (ElasticNet)	—	—	—	—	—	1.000	—	0.0%	100%

The Belgian electricity case shows a balanced split between PM and own-return lags. Under Ridge, after normalizing to sum to 1.0, the PM lags collectively account for approximately 44% of total importance share, with lags 1 and 3 equal at 0.163 and lag 2 at 0.119. The own-return lags carry shares of 0.213 (`BEL_lag1`) and 0.205 (`BEL_lag2`), making them the two largest individual predictors, with the three own-return lags together accounting for about 52% of total importance. The time-to-expiry feature contributes approximately 3.9%. This distribution shows that the nuclear prediction market lags carry meaningful weight in the fitted response — roughly comparable in aggregate to the own-return lags — though the autoregressive structure of the Belgian residual series remains the individually dominant contributor. The PM signal is best characterized as a genuine secondary source of predictive weight rather than a negligible one.

The Argentine ElasticNet case shows near-complete regularization. After normalisation, the four surviving features and their shares are: y_{lag2} (0.411), `tte_current` (0.294), x_{lag2} (0.178), and y_{lag1} (0.117); all other features — including x_{lag1} , x_{lag3} , y_{lag3} , and all remaining TTR lags — are zeroed and therefore appear as 0.000 in the table, confirming that the column sums to 1.000. The PM share (from the single non-zero x coefficient, `x_lag2`) is 17.8%, while the TTR countdown `tte_current` alone accounts for 29.4% of total importance.

This confirms the finding discussed above: the most informative predictor in the 148-day window is the countdown to the election date rather than the direction of LLA prediction market returns. When properly regularized, the model exploits the systematic time trend in MERV returns as election day approaches rather than the PM signal itself.

The AGI case is the most extreme. The only non-zero coefficient is on $y_{\text{lag}3}$ (-0.000516), with every other feature set to exactly zero by regularization. The counterfactual sensitivity to $x_{\text{lag}1}$ is 0.0 at every tested value. The PM lags contribute nothing to the fitted response, and the own-return lags contribute negligibly beyond a single third lag.

6.2.2 When Does the Prediction Market Signal Matter?

The feature-importance comparison suggests that the prediction market signal plays materially different roles across the three cases, and the pattern is more nuanced than a simple story about PM versus own-return dominance.

In the Belgian electricity case, the near-equal split between PM lags and own-return lags under Ridge suggests a genuinely dual-source predictive structure. PM lags 1 and 3 are as important as the own-return lags individually, indicating that the nuclear-detonation prediction market carries persistent lagged information about Belgian idiosyncratic electricity returns across a range of horizons from one to three trading days. This is consistent with a diffuse, multi-channel transmission mechanism: changes in perceived nuclear-detonation risk may propagate through gas futures, interconnection flows, and energy-security positioning over several days rather than being incorporated instantly. In this setting, the framework’s full three-lag feature set is justified, and the PM signal contributes approximately half the model’s total predictive weight.

The Argentine case identifies a different regime. After regularization, the only surviving PM feature is $x_{\text{lag}2}$, and the dominant predictor is the time-to-expiry countdown. This suggests that within the optimal 148-day window, the most reliable predictor of MERV idiosyncratic returns is not the direction of LLA prediction market movements but rather the systematic return pattern as the election date approaches. The finding implies that PM-based hedging in this context should focus less on the level or direction of the prediction market signal and more on the timing of the event itself. The optimal window of 148 days — beginning in approximately late May 2025 — suggests that electoral salience began to be priced into Argentine equities roughly five months before the November election, but only weakly enough that regularization removes the PM lag signal once the time trend is controlled for.

The AGI case identifies a third regime: when the PM is structurally incapable of generating a usable daily-frequency signal — because it is sparse, weakly incentivized, and conceptually distant from the target asset — the PM lags contribute nothing regardless of which model family is used. (Servan-Schreiber et al. 2004).

6.2.3 Implications for Combining Prediction Markets with Other Hedging Signals

The feature-importance results have a direct implication for how the framework developed in this thesis should be positioned relative to existing hedging practice. In the Belgian case,

the PM signal accounts for more than half of the model’s fitted response, suggesting that the PM is a genuine first-order input to the hedge-sizing recommendation rather than a marginal adjustment. In Argentina, the surviving effect is small and mediated through a time trend rather than through the PM price level. In AGI, there is no PM contribution at all.

This heterogeneity suggests that the framework’s value is contingent on the specific asset-event pair and the structural conditions of the PM, as summarized in Section 6.4.1 below. When those conditions are met, the PM signal can account for a majority of the predictive input and generate a material hedge demand through the CVaR augmentation even when point predictions are modest. When they are not met, the framework correctly returns a weak or zero recommendation, which is itself informative.

Several natural extensions follow. First, the scenario probabilities generated by the branching simulation could be used to scale an existing hedge notional rather than initiating one from scratch, consistent with the use of PM signals as probability-weighted adjustments to scenario-based risk frameworks (Snowberg, J. Wolfers, and E. Zitzewitz 2008; Goodell, McGee, and McGroarty 2020). Second, the salience-window finding from Argentina suggests that PM-informed hedging is best deployed as a short-duration overlay activated around identifiable event windows, consistent with the option-overlay literature (IsraelovNielsen2014; Hull 2018). Third, the Belgian result demonstrates that a PM share exceeding 50% of total model importance is achievable even under regularized linear estimation, and that this level of PM contribution is sufficient to generate a material CVaR-informed hedge demand even when the point-forecast OOS R^2 is negative.

6.3 Limitations of the Framework

The three case studies collectively demonstrate that the proposed framework can produce useful scenario-based outputs when the structural conditions are favorable, but they also expose a set of limitations that constrain its reliability and generalizability. These limitations span data quality, market design, empirical evaluation, and implementation.

6.3.1 Data Frequency and the Timing Problem

The most pervasive data limitation in the thesis is the mismatch between the temporal resolution at which the relevant financial information arrives and the daily frequency at which the analysis is conducted.

In the Argentine case, the event of interest has a known resolution date and a well-defined window of heightened political salience in the weeks before the vote. Financial markets are known to incorporate political information rapidly, often within hours or even minutes of its arrival (Bauer and T. M. Mertens 2023; Engelberg, Reed, and Ringgenberg 2011). The daily aggregation of both equity returns and prediction market prices necessarily smooths over these intraday dynamics. Any relationship between electoral expectations and Argentine equity returns that operates at sub-daily frequency will be attenuated or entirely lost in a daily-frequency model. High-frequency event-window methods are specifically designed for this setting (Bauer and T. M. Mertens 2023).

The Belgian electricity case faces a related but structurally different frequency problem. Wholesale electricity prices settle at quarter-hourly intervals, and intraday scarcity dynamics

can produce large price spikes that are smoothed away by daily averaging. The influence diagnostics for the factor model identify 5 October 2025 as an extreme outlier with a Cook’s distance of 0.972 and a studentised residual of 136, almost certainly reflecting an intraday market stress episode. The decision to work at daily frequency was driven by the availability of prediction market data, which was accessible only as a daily series. This constraint should be treated as a data acquisition problem rather than a methodological choice.

6.3.2 Play-Money and Incentive-Weak Prediction Markets

The AGI case introduces a different category of data limitation: the quality of the prediction market signal itself. The AGI-expectations measure is constructed from forecast aggregations on Metaculus and Manifold Markets, both of which use scoring rules or play-money tokens rather than real financial stakes.

The theoretical case for PM efficiency rests on the incentive to acquire and reveal private information when doing so is financially rewarded (Hayek 1945; Justin Wolfers and Eric Zitzewitz 2004). When the financial stake is removed, the incentive to incur the cost of gathering precise information is weakened. The empirical evidence is mixed (Servan-Schreiber et al. 2004), but the more salient concern for this thesis is update speed: even if play-money markets converge to accurate probability estimates over time, they may do so more slowly than cash markets, rendering their lagged values uninformative for next-day forecasting. The cross-correlation analysis, which finds negligible CCF values at all lags from zero to twenty days, cannot distinguish between “no genuine relationship exists” and “the lag structure is too uncertain to estimate from a daily series.”

A more fundamental concern is the conceptual distance between the measured variable and the financial asset. AGI-timeline questions measure multi-year horizon shifts that transmit only indirectly into daily semiconductor and AI-infrastructure equity returns. The null result in the AGI case should therefore be interpreted partly as evidence that the chosen proxy is too conceptually distant from the target financial outcome to generate a detectable daily-frequency relationship.

6.3.3 Absence of a Closed-Loop Backtest of Hedge Effectiveness

The most important empirical limitation of the thesis is that it does not close the loop between the hedge-sizing recommendation and a realized assessment of whether that recommendation would have been effective. In all three case studies, the hedge-sizing routine operates at a single forecast origin, producing one terminal recommendation rather than a rolling time series of hedge signals. No rolling history of triggered and untriggered hedge recommendations exists, and the analysis cannot assess Type I errors (the hedge fires unnecessarily), Type II errors (the hedge fails to fire when a loss materialises), or the realized hedge efficiency conditional on activation.

This limitation cannot be easily remedied with the available data. Conducting a meaningful closed-loop backtest would require a sample containing multiple distinct episodes in which the PM signal moves materially, the hedge is triggered, and the subsequent financial outcome is observable. Given sample lengths of one year or less, and given the episodic nature of PM movements, the number of independent episodes is too small to support statistically

reliable inferences (Bailey et al. 2014). A related concern applies to the Belgian electricity result specifically: the 44% PM feature-importance share is estimated from a single year of daily data, raising the possibility of a spurious correlation. A one-year period is long enough for coincidental co-movement between nuclear-PM returns and Belgian electricity residuals to produce a non-trivial regularized coefficient, but too short to distinguish a structural relationship from a data-specific artefact. Replication across multiple years, or across multiple conflict-linked PM contracts and electricity markets, would be necessary before treating the Belgian Ridge result as evidence of a robust, economically meaningful predictive link rather than a 2025-specific finding.

The hedge-sizing parameters — execution haircut of 0.70 and hedge-efficiency parameter of 0.60 — are stylised assumptions rather than empirically estimated quantities. More rigorous implementation would require calibrating both from historical execution data or option pricing models. And the hedge recommendation is sensitive to the exposure convention: the sign of the investor’s exposure must be correctly specified before the framework produces meaningful output.

6.3.4 Liquidity, Execution, and the Gap between Signal and Trade

Even setting aside the backtest limitations, there is a structural gap between the hedge-sizing signal produced by the framework and an implementable trading strategy.

On the PM side, approximately 63% of short-term Polymarket contracts exhibit zero trading volume over a 24-hour period, and more than half have less than \$100 in available liquidity (Panewslab 2023; Majumder et al. 2009). The nuclear-detonation contract used in the Belgian electricity case was removed from Polymarket in March 2026 following regulatory pressure and the DEATH BETs Act proposal (Ostroff and Osipovich 2026; U.S. Congress 2026). Belgium’s own legal framework prohibits PMs entirely (Marjanović 2025), meaning a Belgian energy participant could not use the Polymarket contract as a direct hedge instrument regardless of its informational value.

The practical implication, acknowledged in the methodology chapter, is that the prediction market series functions primarily as an information mechanism rather than a direct hedge vehicle. The actual hedge would need to be implemented through a derivative overlay on the underlying financial asset — put options, collars, or futures on Belgian electricity or natural gas — introducing basis risk between the PM-implied scenario distribution and the available standardized instruments (Axén and Cortis 2020; Hull 2018). The hedge notional recommendations should be understood as upper bounds on implementable coverage rather than executable instructions (Almgren and Chriss 2001; Perold 1988).

6.4 Practical Value and Directions for Future Research

Despite the limitations identified in the previous section, the three case studies collectively establish that the framework has practical value in identifiable conditions, and they point toward a productive research agenda.

6.4.1 Conditions for Framework Utility

The cross-case evidence suggests that the framework produces actionable output when four structural conditions are jointly satisfied. First, the prediction market must be cash-settled and actively traded. Second, the financial asset must have a plausible and theoretically grounded exposure to the event being traded: the channel through which the event affects the asset should be identifiable, even if indirect, as in the Belgian electricity case where the nuclear-risk signal transmits through gas markets and energy-policy expectations. Third, the data frequency must be matched to the speed at which event information is incorporated into prices. Fourth, the sample must cover a period of meaningful variation in the event probability. When all four conditions are met, the framework produces separated scenario branches, a material CVaR-informed hedge demand, and feature-importance rankings that confirm a genuine role for the PM signal. When one or more conditions fail, the framework correctly produces a weak or null recommendation, which is itself informative.

6.4.2 Prediction Markets as Saliency Detectors

One of the more practically interesting findings to emerge from the Argentine case is the strong sensitivity of model performance to the length of the training window. As shown in the case-study chapter, ElasticNet achieves its best window-specific RMSE of 0.019 and OOS R^2 of +0.300 on a 148-day training window, whereas the same model trained on the full 168-day history produces a negative OOS R^2 . This improvement is not unique to ElasticNet: Ridge achieves $R^2 = +0.295$ on a 150-day window, and the cluster of best-performing specifications is concentrated in windows of 148–163 days. By contrast, the VECM achieves its best performance on a 144-day window at OOS $R^2 = +0.277$ — still positive, but below the leading ML specifications.

This window-sensitivity pattern carries an interpretation that merits careful framing. The feature-importance analysis in Section 6.2.1 shows that within the optimal 148-day window, the ElasticNet zeroes out all LLA prediction market lags; the dominant predictor is `tte_current`, the countdown to the election date. The positive OOS R^2 of +0.300 is therefore driven primarily by the systematic time trend in MERV returns as election day approaches, not by the directional content of the LLA PM signal. The Argentine finding is thus better described as an *election proximity detector* than a prediction market saliency window: the model learns that Argentine equities exhibit a systematic momentum pattern in the final five months before the election, regardless of whether the LLA contract is moving up or down on any given day. The prediction market lags themselves provide only a small secondary contribution (17.8% PM share after normalisation, with only x_{lag2} surviving regularization).

That said, the window-sensitivity result still has practical value. The contraction of the optimal window to approximately 148–150 days — corresponding to a start date around late May 2025, roughly five months before the November election — marks the onset of the systematic pre-election return pattern. A researcher monitoring the optimal window in real time could use its stabilisation at this length as an indicator that election-proximity dynamics are being priced and that a review of electoral-hedge positioning is warranted, even if the PM signal itself is not the operative predictor. Future work should explicitly test whether the time-trend predictor can be separated from the PM signal by constructing counterfactual

models that include `tte_current` but exclude the PM lags, to confirm whether the window-sensitivity result is entirely attributable to the time trend rather than to any residual PM information content (Justin Wolfers and Eric Zitzewitz 2004; Snowberg, J. Wolfers, and E. Zitzewitz 2008).

6.4.3 Directions for Future Research

Four directions emerge directly from the limitations and findings of this thesis.

Higher-frequency data and intraday event windows. The most immediate empirical extension is to replicate the framework at intraday frequency using tick-level PM data alongside high-frequency financial returns. Applying the same factor residualisation and machine-learning pipeline to intraday data would sharpen the identification of lead-lag relationships, reduce the temporal aggregation bias that attenuates the daily-frequency results, and make the scenario simulation more relevant for short-horizon hedging decisions. For the Argentine case specifically, an event-window study centred on the days immediately surrounding major polling releases or debate events would be a natural starting point, following the methods developed in Bauer and T. M. Mertens (2023) and Engelberg, Reed, and Ringgenberg (2011).

Expanding the asset-event pair library. The three case studies represent only a narrow slice of the potential application space. Future work could systematically evaluate the framework across a larger library of asset-event pairs, including sovereign bond spreads and political risk markets, commodity prices and weather or supply disruption contracts, and individual equity exposures and product-approval or regulatory-outcome markets. A panel approach across many such pairs would allow empirical regularities to be identified: for instance, whether the optimal training window consistently shortens as events approach, whether the PM importance share is correlated with the liquidity or incentive structure of the market, and whether the scenario-branch separation is predictable from observable market microstructure variables such as open interest or bid-ask spreads.

Closed-loop hedge effectiveness evaluation. As discussed in Section 6.3.3, the absence of a rolling backtest of hedge recommendations is the most significant evaluative gap in the thesis. Addressing this requires either longer historical samples — which may become available as PMs mature — or simulation-based approaches in which the true event probability path is generated synthetically and the framework’s recommendations are evaluated against that path. Bailey et al. (2014)’s procedures for guarding against backtest overfitting provide a natural methodological template. Such an exercise would also enable calibration of the risk-budget threshold and execution haircut parameters, which are currently set heuristically.

Integration with derivative pricing for executable recommendations. The current framework stops at the notional coverage recommendation and does not attempt to price the instruments through which that coverage would be implemented. Integrating the scenario-based downside measure with a standard option-pricing model — for instance, mapping the

CAR distribution implied by the branching tree onto a strike and tenor for a put option or collar — would close this gap and allow the framework to produce cost-aware hedging recommendations rather than purely notional ones (**israelovNielsen2014**; Hull 2018). For the Belgian electricity case, where standardized derivatives on wholesale electricity and natural gas are available through organized exchanges, such an extension would be practically feasible with publicly available options data.

7 Conclusion

7.1 What This Thesis Set Out to Do

This thesis asked a simple question with a complicated answer: can event-linked prediction markets be incorporated into an empirical risk-management framework for traded securities? The question was motivated by an observable gap between the informational role that PMs are theoretically capable of playing and their near-total absence from systematic hedging practice. The gap is not merely academic. As the introduction documented, the same contracts that regulators and legislators have sought to prohibit on moral grounds — contracts on nuclear detonation, conflict, terrorism, and war — are precisely the contracts most plausibly linked to assets whose investors face material event-driven exposures. Understanding whether those contracts carry usable hedging information is therefore both an empirical question and a policy-relevant one.

7.2 The Methodological Contribution

Before turning to what the case studies found, it is worth stating clearly what this thesis contributes at the level of methodology, independently of the empirical results. The framework developed here is a complete, modular pipeline: factor residualization to isolate the idiosyncratic return component, a bakeoff-plus-validation model selection procedure that respects temporal ordering throughout, a branching scenario simulation calibrated to the empirical distribution of PM movements, and a dual-output hedge-sizing routine that reports both a point-estimate notional and a CVaR-augmented bootstrap notional. Each stage is designed to address a specific failure mode that a naive implementation would overlook: look-ahead bias in factor estimation, in-sample overfitting in model selection, equal-probability assumption errors in branching, and the tendency to confuse scenario-distribution risk with innovation-distribution risk in the CVaR stage.

These design choices are not incidental. They ensure that the outputs of the framework — whether a €800,000 hedge recommendation or a zero notional — are interpretable and defensible rather than artifacts of methodological shortcuts. The pipeline generalizes across asset classes, PM contract types, and event horizons, requiring only that a cash-settled PM with adequate trading history exists for the event of interest. Whether applied to a conflict-linked energy market, a political election, or a technology timeline, the same sequence of steps produces outputs whose provenance can be traced and whose limitations can be stated precisely.

7.3 The Gradient of Conditions

The three case studies were not selected to confirm the framework but to stress-test it across materially different conditions. What they reveal collectively is a structured gradient rather than a single answer to the central question.

The Belgian electricity case is the strongest. A Ridge regression model trained on lagged nuclear-detonation PM returns and Belgian idiosyncratic electricity returns assigns approximately 44% of its normalized importance weight to the PM lags — a share large enough

to be substantively meaningful even under conservative interpretation. The 20-day rolling CVaR_{80} co-moves episodically with PM return volatility over the 2025 sample, suggesting that geopolitical risk episodes and electricity-price tail risk cluster in time even if the lead-lag relationship is weak at average daily frequency. A hedge notional of €800,000 is triggered on both the point-estimate and bootstrap bases under the short-price consumer exposure. The result is not a strong causal finding — a single year of data and a negative expanding-window OOS R^2 make any causal claim premature — but it is a coherent empirical signal: the nuclear PM carries information about Belgian electricity residuals that is captured in a regularized linear model and translates into a concrete hedge recommendation.

The Argentine case is informative at a different level. The ElasticNet model zeroes out all LLA prediction market lags in its optimal 148-day window, assigning the dominant predictive role to the time-to-expiry countdown. The positive in-window OOS R^2 of +0.300 reflects systematic pre-election return momentum in Argentine equities rather than PM-directional signal. The hedge notional is zero. At face value this is a null result, but its structure reveals something important: the relationship between event expectations and equity returns is not stable across the year, and the window in which it becomes measurable — the five months before the election — is identifiable without prior knowledge of the election date. An investor monitoring the optimal training-window length in real time would have observed this window contracting as the election approached, providing a signal that PM-informed hedging was becoming worth considering, even if the PM lags themselves were ultimately dominated by the time trend. This is the sense in which the Argentine case is an *election proximity detector* rather than a PM signal detector.

The AGI case closes the gradient at the null end. Six independent diagnostics — the CCF analysis, the ARDL selection, the Local Projections IRF, the ARCH test, the feature importance audit, and the counterfactual sensitivity sweep — all confirm the same conclusion: the AGI signal constructed from play-money forecasting platforms carries no detectable daily-frequency information about ARTY idiosyncratic returns. The CVaR bootstrap triggers a €800,000 notional, but only because ARTY’s residual distribution is heavy-tailed; remove the AGI signal entirely and the CVaR fires identically. This is not a failure of the framework; it is the framework working as intended. A methodology that correctly identifies a null result and explains why, using a systematic diagnostic battery, is more valuable than one that returns a non-zero hedge recommendation regardless of data quality. The AGI case delineates the boundary conditions precisely: when the PM is weakly incentivized, structurally sparse, and conceptually distant from the asset, the appropriate output is zero.

Taken together, the three cases map a clear set of conditions for framework utility. The PM must be cash-settled and actively traded; the transmission channel from the event to the asset must be theoretically identifiable; the data frequency must match the speed at which event information is incorporated into prices; and the sample must contain genuine variation in the event probability. When those conditions are met, the framework produces separated scenario branches, a coherent hedge-sizing signal, and feature-importance rankings consistent with a genuine PM contribution. When they are not, it correctly produces a weak or null recommendation.

7.4 What Would Strengthen These Findings

The most important empirical limitation is one that cannot be resolved within the scope of this thesis but should be named directly. All three case studies rest on a single year of daily data. For the Belgian electricity result in particular, a 44% PM importance share and a triggered hedge notional are consistent with a genuine structural relationship — but they are also consistent with a 2025-specific spurious correlation between nuclear-detonation contract movements and Belgian electricity residuals. Distinguishing between these explanations requires replication: across multiple years, across multiple conflict-linked PM contracts, and ideally across multiple European electricity markets with different nuclear exposure profiles. Until such replication exists, the Belgian result should be treated as a well-grounded hypothesis rather than an established finding.

A second important limitation is data frequency. The analysis is conducted at the daily level throughout, forced by the granularity of available PM price histories. There is good reason to think this misses much of the relevant variation. Financial markets are known to incorporate political and geopolitical information within hours or even minutes of its arrival (Bauer and T. M. Mertens 2023; Engelberg, Reed, and Ringgenberg 2011), while daily aggregation smooths over exactly the intraday dynamics that would sharpen identification. The Argentine case in particular — where directional accuracy struggles across all model families — may look materially different at hourly frequency in a narrow window around key polling releases or campaign events. Several platforms now provide API access to sub-daily PM price histories for actively traded contracts. Future work should treat higher-frequency PM data not as a refinement but as the primary empirical target, with the daily results serving as an accessible baseline.

7.5 Closing: The Case for Legitimate Use

This thesis began with a controversy. In early 2026, the DEATH BETs Act was proposed in the United States Senate, seeking to prohibit prediction market contracts on terrorism, assassination, war, and related violent events. Supporters of the bill argued that there is no legitimate public benefit in allowing markets that appear to profit from human suffering. Shortly thereafter, Polymarket closed its contract on nuclear weapon detonation — one of the three contracts at the center of this analysis — partly in response to that same regulatory and reputational pressure.

The question this thesis has tried to answer is whether those supporters are right. Not morally, which is beyond the scope of empirical finance, but empirically: do such contracts serve a legitimate risk-management function for agents with genuine exposures to the events they price?

The answer from the Belgian electricity case is that they can. A market participant exposed to Belgian wholesale electricity prices in 2025 — a gas-intensive industrial consumer, an energy retailer, or a utility with unhedged forward exposure — faced a year in which geopolitical tail risk was a first-order driver of the price environment. The nuclear-detonation contract on Polymarket, during the period it was available, carried information about that risk environment that was not captured by weather, load, or conventional market fundamentals alone. The framework developed in this thesis would have directed that

participant toward a €800,000 hedge notional on their electricity exposure — a recommendation grounded not in speculation about nuclear events but in the documented co-movement between shifts in perceived geopolitical risk and electricity-price tail dynamics.

That is what legitimate use looks like. It does not require that the underlying event be likely, or that the PM contract itself be used as the hedge instrument — Belgium’s own legal framework would prohibit that in any case. It requires only that the contract aggregates information about a risk that matters for real economic agents, and that a systematic methodology exists for translating that aggregated information into defensible portfolio decisions. This thesis has demonstrated that both conditions are satisfiable in at least one empirically documented setting.

The debate over prediction markets and their regulation will continue. Reasonable people disagree about where to draw the line between socially useful risk aggregation and morally objectionable speculation on catastrophe. That debate will not be settled here. But it should be conducted with the empirical record in view. Prohibiting contracts on violent events without considering their informational and risk-management value is analogous to prohibiting short selling without considering its contribution to price discovery — an instinct that is understandable but that may impose real costs on market participants who would otherwise use those instruments prudently. The evidence in this thesis suggests that the costs of prohibition are not zero, and that building the empirical case for legitimate use is a more productive response to regulatory pressure than conceding the field.

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